

Kathmandu Bernhardt College



Laboratory Manual

Digital Logic Design
(CSC-116)

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1. Guide to Assembling your Circuits:

In this section we describe the use of the breadboard and give basic hints about the wiring process needed to power up and interconnect your circuits.

Assembling circuits on your breadboard is a fast and easy process once you get used to it. To assemble your circuit first select the chips that you need, insert them in the breadboard, wire up the power and ground connections as described in the next section and next wire the logic elements according to the circuit connections that you obtained from the design process.

Before you insert a chip into the breadboard, make sure it is properly oriented and that when you press it down the pins of the chip actually enter the holes and do not bend underneath the chip package.

When wiring, be careful to hit the right hole needed in the connection, because this is one of the most common mistakes found to cause an error in your projects.

2. Breadboard Description:

In order to build the circuit, a digital design kit that contains a power supply, switches for input, light emitting diodes (LEDs), and a breadboard will be used. Make sure to follow your instructor's safety instructions when assembling, debugging, and observing your circuit. You may also need other items for your lab such as: logic chips, wire, wire cutters, a transistor, etc. Figure 1 shows a common breadboard, while Figure 2 shows how each set of pins are tied together electronically.

For these labs, the highest voltage used in your designs will be five volts or +5V and the lowest will be 0V or ground.

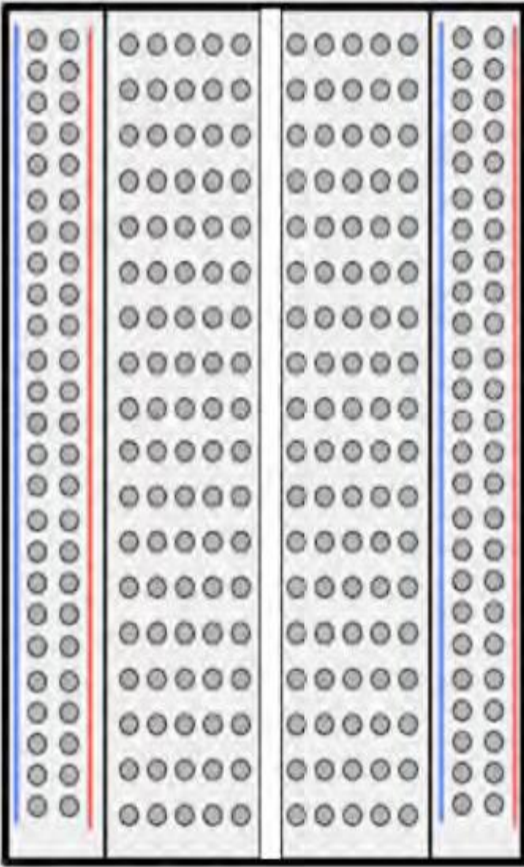


Figure 1: Breadboard

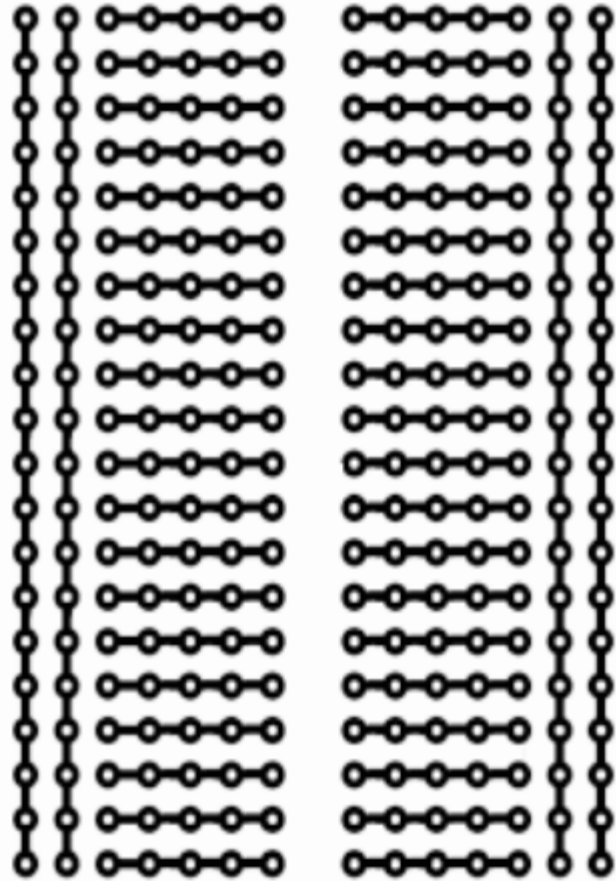


Figure 2 Common connections

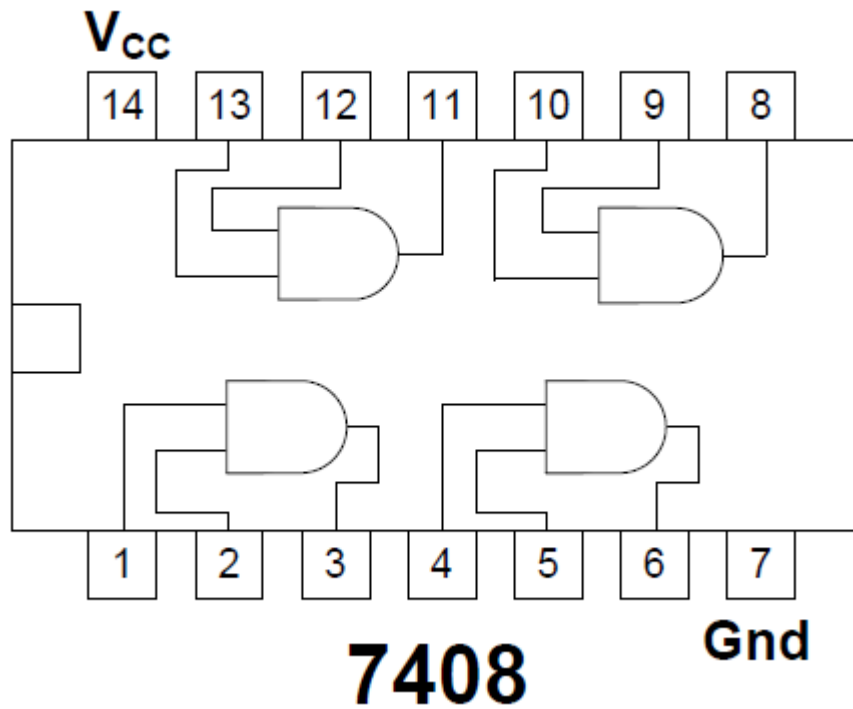
3. Common Causes of Problems:

- Not connecting the ground and/or power pins for all chips.
- Not turning on the power supply before checking the operation of the circuit.
- Leaving out wires.
- Plugging wires into the wrong holes.
- Driving a single gate input with the outputs of two or more gates.
- Modifying the circuit with the power on.

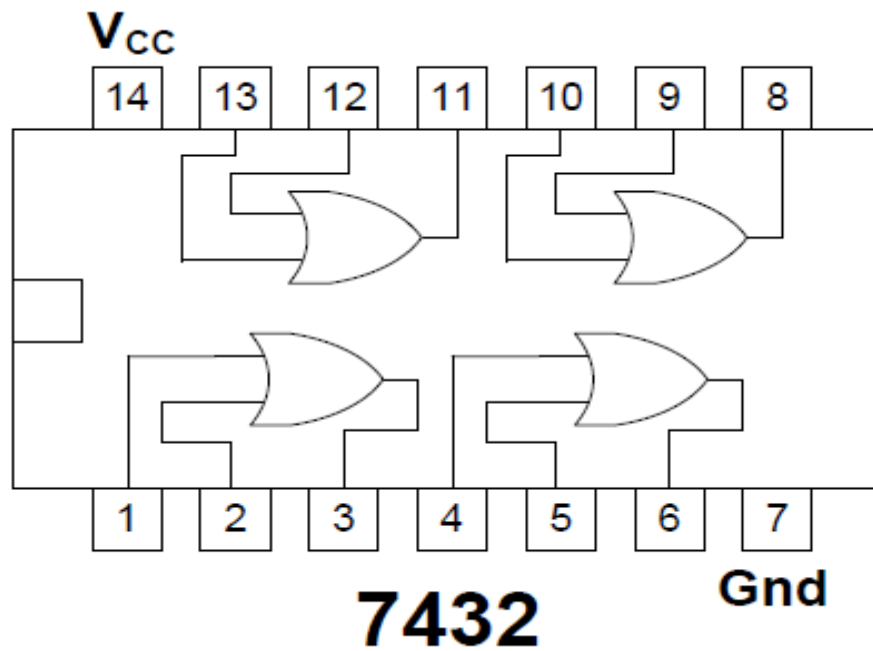
Note: In all experiments, you will be expected to obtain all instruments, leads, components at the start of the experiment and return them to their proper place after you have finished the experiment.

4. Pin diagram of all gates:

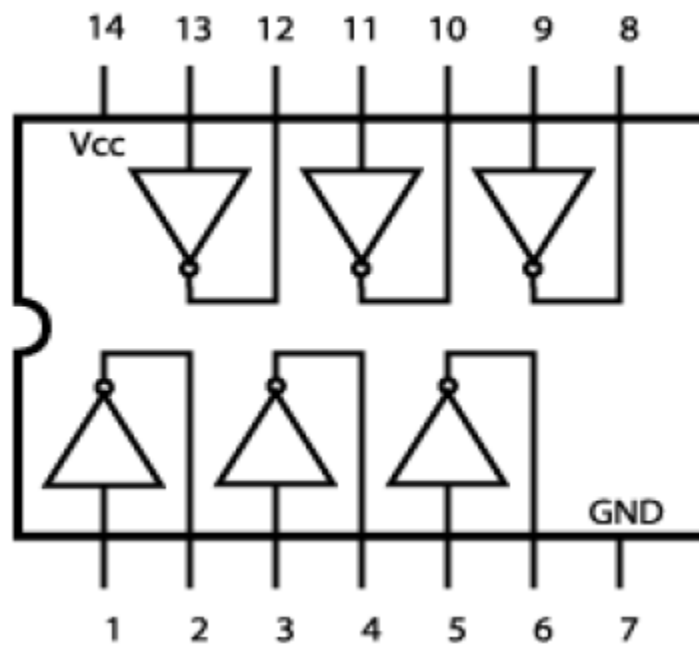
a. AND GATE 7408 (Quad 2 input)



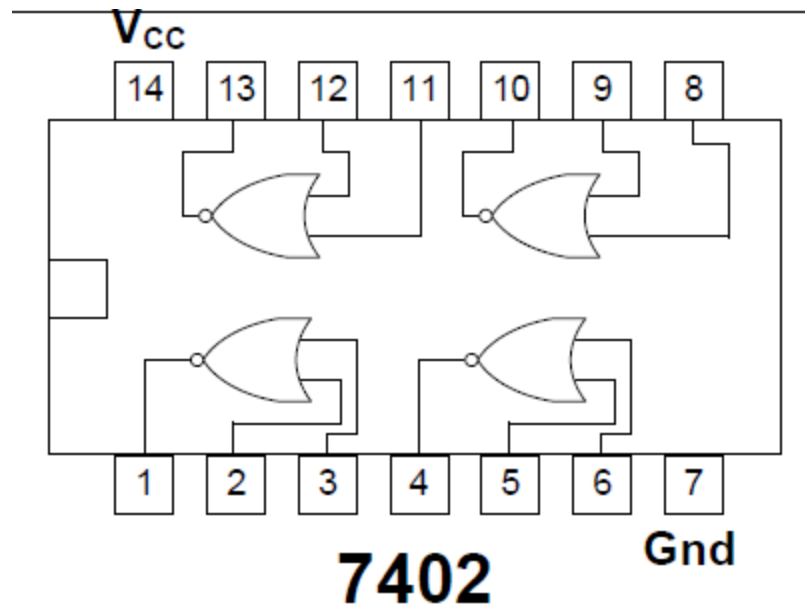
b. OR GATE 7432 (Quad 2 input)



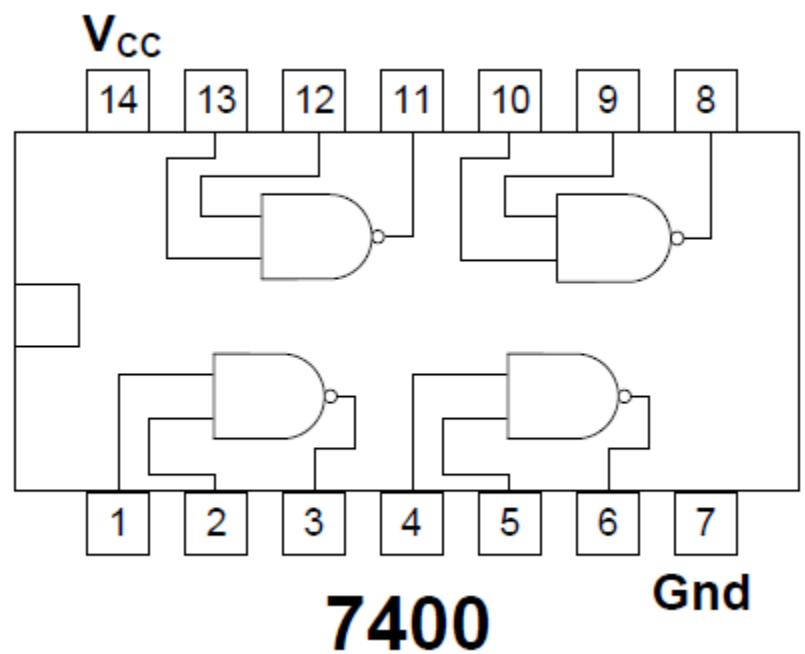
c. NOT GATE 7404



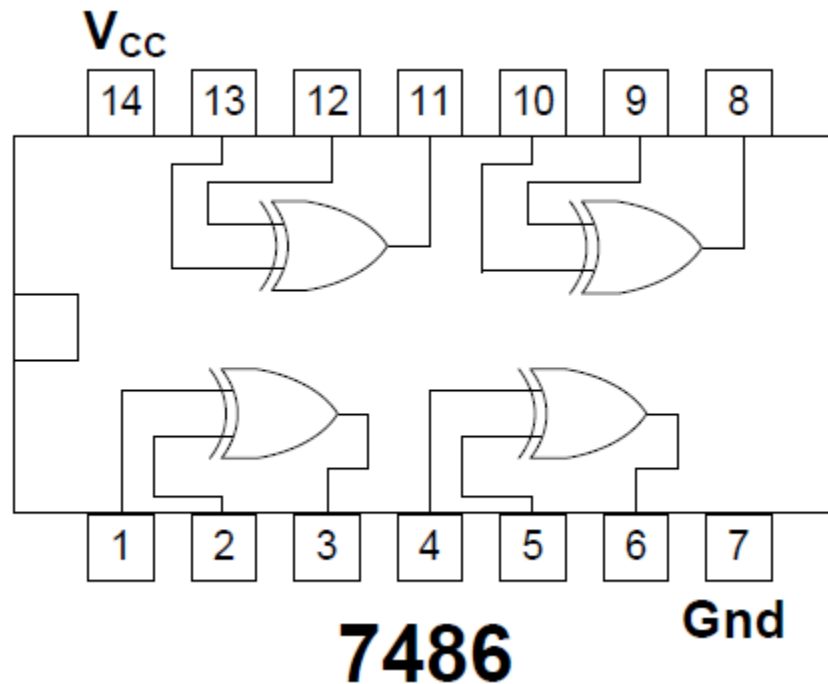
d. NOR GATE 7402 (Quad 2 input)



e. NAND GATE 7400 (Quad 2 input)

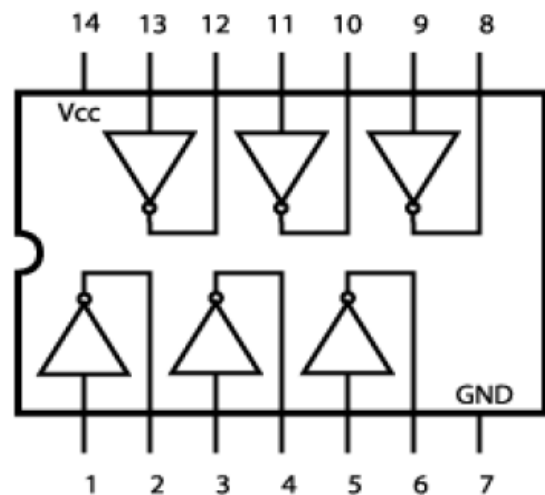
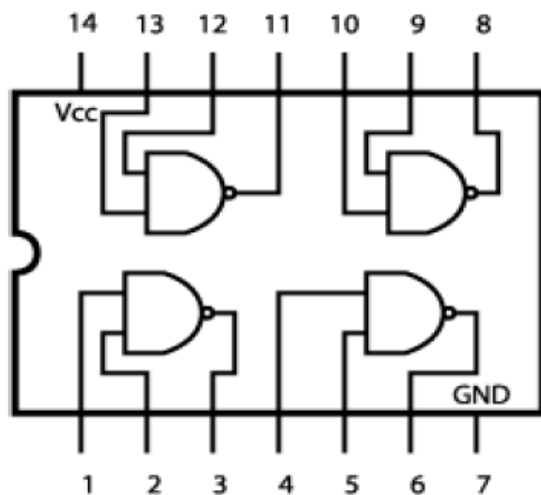


f. EXCLUSIVE XOR GATE 7486 (Quad 2 input)



5. Example Implementation of a Logic Circuit

Build a circuit to implement the Boolean function $F = A \cdot B$ using TTL IC 74LS00 (NAND gate) and TTL IC 7404 (INVERTER) as per discussed in figure.



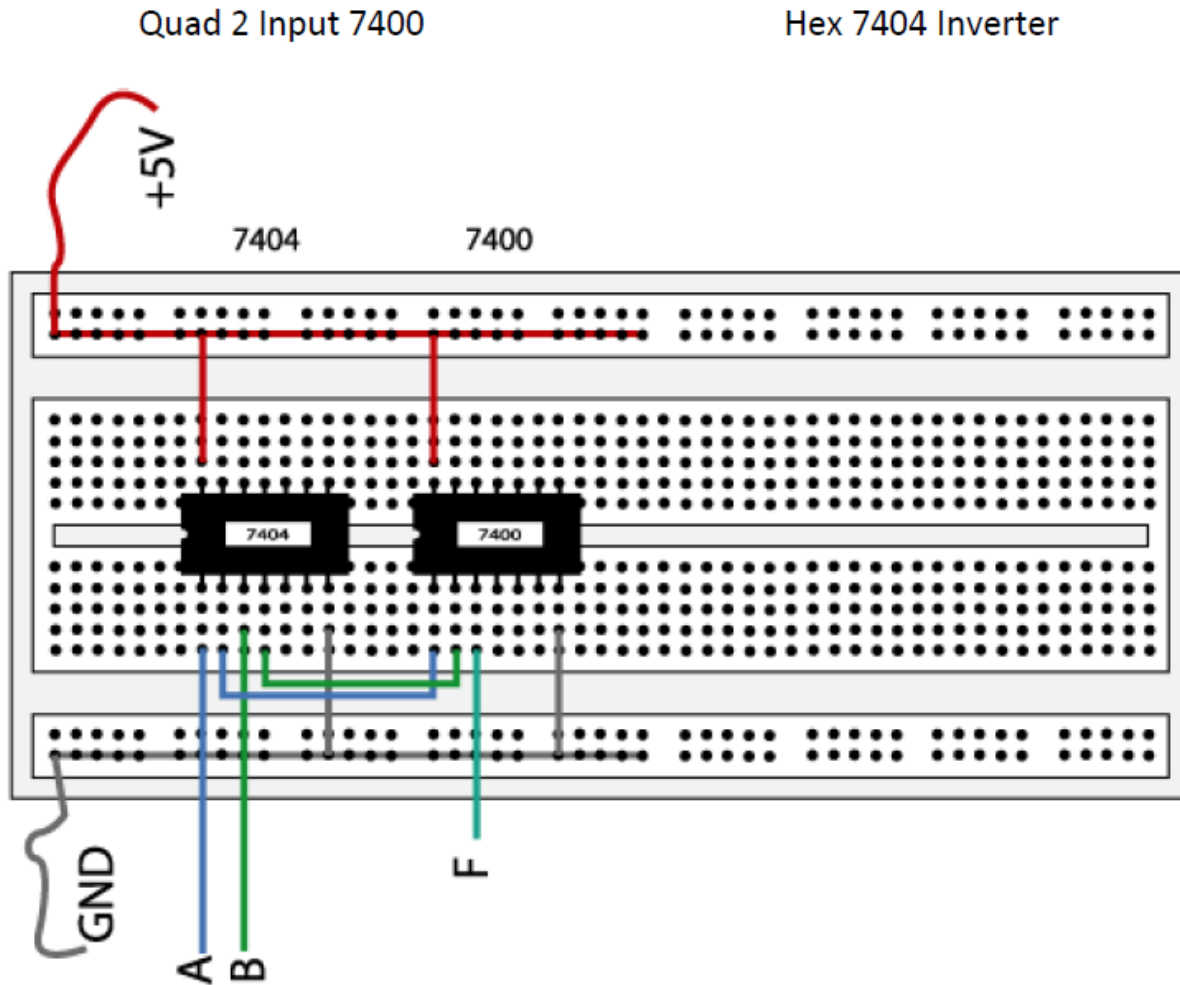


Figure: The complete designed and connected circuit

EXPERIMENT NO: 1

Aim: - Introduction to digital electronics Lab-nomenclature of digital ICs, specifications, study of the data sheet, concept of VCC and Ground, verification of truth tables of logic gates using TTL ICs.

APPARATUS REQUIRED: Power Supply, Digital Trainer Kit., Connecting Leads, and IC's.

Gates	IC NO.
AND	7408
OR	7432
NAND	7400
NOR	7402
NOT	7404
XOR	74136

THEORY: Explain Theory.

PROCEDURE:

- Fix the IC's on breadboard & give the supply.
- Connect the +ve terminal of supply to pin 14 & -ve to pin 7.
- Give input at pin 1, 2 & take output from pin 3. It is same for all except NOT & NOR IC.
- For NOR, pin 1 is output & pin 2&3 are inputs.
- For NOT, pin 1 is input & pin 2 is output.
- Note the values of output for different combination of inputs & draw the TRUTH TABLE.

RESULT: All gates are verified. Observed output matches theoretical concepts.

PRECAUTIONS:

- Make the connections according to the IC pin diagram.
- The connections should be tight.
- The Vcc and ground should be applied carefully at the specified pin only.
- While making connections main voltage should be kept switched off.
- Never touch live and naked wires.

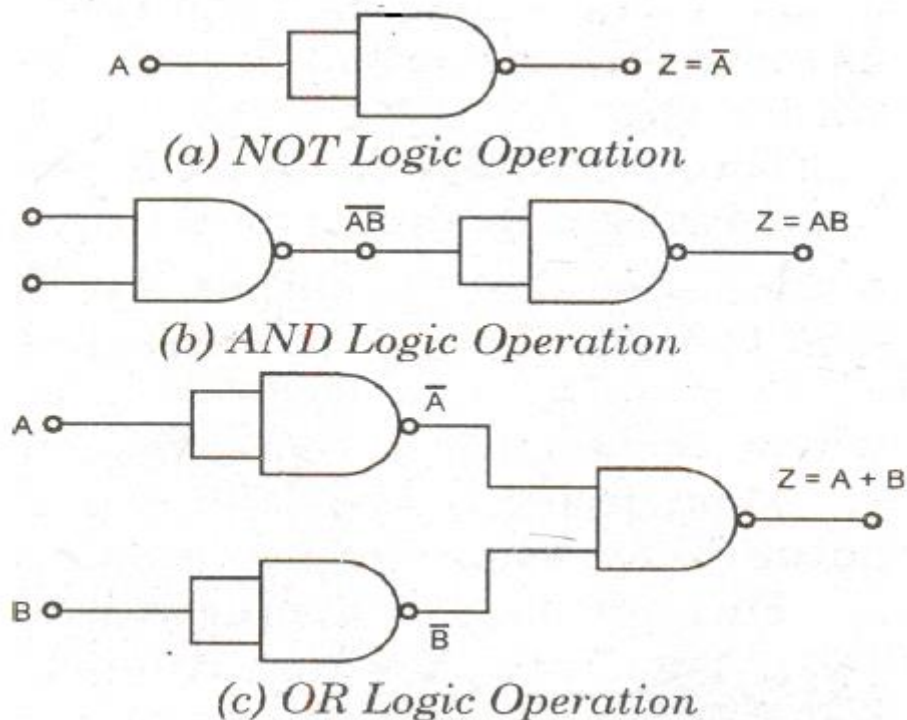
EXPERIMENT NO: 2

Aim: - To study and verify NAND and NOR gate as universal gate.

APPARATUS REQUIRED – Digital trainer kit, IC 7400 (NAND gate).

THEORY – NAND or NOR sufficient for the realization of any logic expression, because of this reason, NAND and NOR gates are known as UNIVERSAL gates.

LOGIC DIAGRAM –



PROCEDURE –

- Make the connections as per the logic diagram
- Connect +5v to pin 14 & ground to pin 7
- Apply diff combinations of inputs to the i/p terminals.
- Note o/p for NAND as universal gate.
- Verify the truth table.

RESULT: All Universal gates are verified. Observed output matches theoretical concepts.

EXPERIMENT NO: 3

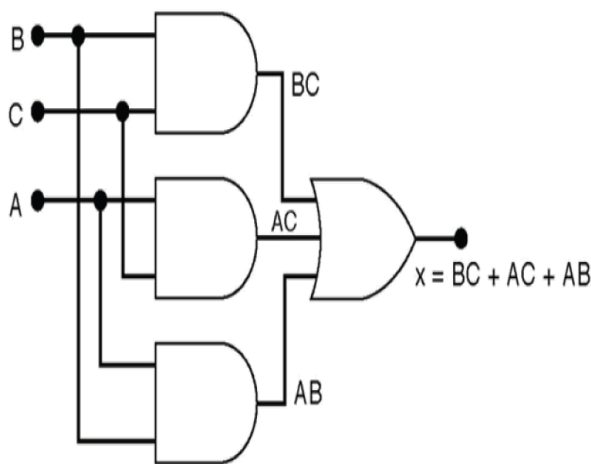
Aim: - Implementation of the given Boolean function using logic gates in both POS and SOP forms.

APPARATUS REQUIRED: Power Supply, Digital Trainer, IC's (7404, 7408, 7432) Connecting leads.

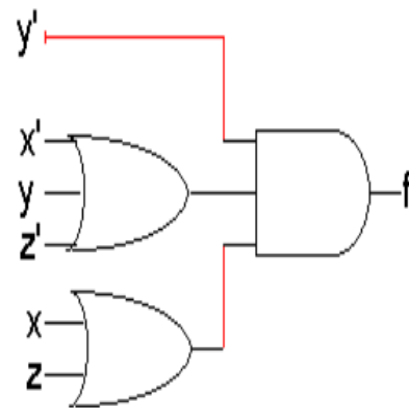
THEORY: Karnaugh maps are perhaps the most extensively used tool for simplification of Boolean functions. It is mostly used for functions having up to six variables beyond which it becomes very cumbersome. In an n-variable K-map there are 2^n cells. Each cell corresponds to one of the combination of n variable, since there are 2^n combinations of n-variables.

Example- $f(A,B,C,D)=A'BC+AB'C+ABC'+ABC$ (SOP) Reduced form is $BC+AC+AB$, and POS form is $f(X,Y,Z)=Y'(X'+Y+Z')(X+Z)$.

CIRCUIT DIAGRAM:



SOP Form



POS Form

PROCEDURE:

- With given equation in SOP/POS form first of all draw a K-map.
- Enter the values of the O/P variable in each cell corresponding to its Min/Max term.
- Make group of adjacent ones.
- From group write the minimized equation.

e) Design the ckt. of minimized equation & verify the truth table.

RESULT:

Implementation of SOP and POS form is obtained with AND and OR gates.

EXPERIMENT NO: 4

Aim: - To design and verify operation of half adder and full adder.

APPARATUS REQUIRED: Power supply, IC's (7486, 7432 and 7408), Digital Trainer, Connecting leads.

THEORY: We are familiar with ALU, which performs all arithmetic and logic operation but ALU doesn't perform/ process decimal no's. They process binary no's.

Half Adder: It is a logic circuit that adds two bits. It produces the O/P, sum & carry. The Boolean equation for sum & carry are

$$\text{SUM} = A \oplus B$$

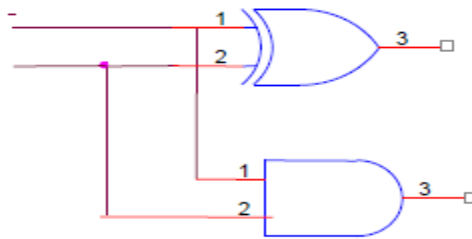
$$\text{CARRY} = A \cdot B$$

Therefore, sum produces 1 when A&B are different and carry is 1 when A&B are 1. Application of Half adder is limited.

Truth Table:

Truth Table			
Input		Output	
A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

CIRCUIT DAIGRAM:



Full Adder: It is a logic circuit that can add three bits. It produces two O/P sum & carry. The Boolean Equation for sum & carry are

$$\text{SUM} = (A \text{ XOR } B) \text{ XOR } C_{in} = (A \oplus B) \oplus C_{in}$$

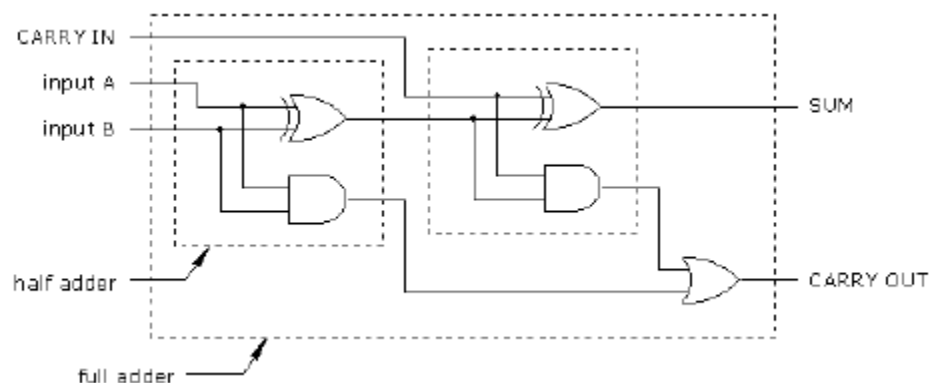
$$\text{CARRY} = A \text{ AND } B \text{ OR } C_{in}(A \text{ XOR } B) = A.B + C_{in}(A \oplus B)$$

Therefore, sum produces one when I/P is containing odd no's of one & carry is one when there are two or more one in I/P.

Truth Table:

Input			Output	
A	B	Cin	Sum	Carry
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

CIRCUIT DAIGRAM:



RESULT: The Half Adder & Full Adder ckts. are verified.

EXPERIMENT NO: 5

Aim: - To design and verify operation of half subtractor and full subtractor.

APPARATUS REQUIRED: – Digital trainer kit,
IC 7486 (EX-OR),
IC 7408 (AND gate),
IC 7404 (NOT gate).

THEORY:

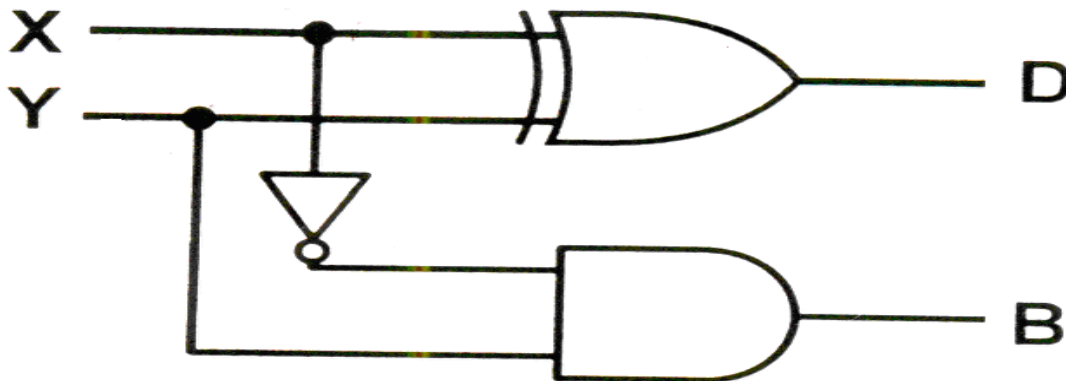
The half-subtractor is a combinational circuit which is used to perform subtraction of two bits. It has two inputs, X (minuend) and Y (subtrahend) and two outputs Difference and Borrow.

The Boolean expression:

$$\text{Difference} = A \oplus B$$

$$\text{Borrow} = A' B$$

CIRCUIT DAIGRAM:



Truth Table:

Inputs		Outputs	
A	B	Difference	Borrow
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

A full subtractor is a combinational circuit that performs subtraction involving three bits, namely A (minuend), B (subtrahend), and B_{in} (borrow-in). It accepts three inputs: A (minuend), B (subtrahend) and a B_{in} (borrow bit) and it produces two outputs: D (difference) and Bout (borrow out).

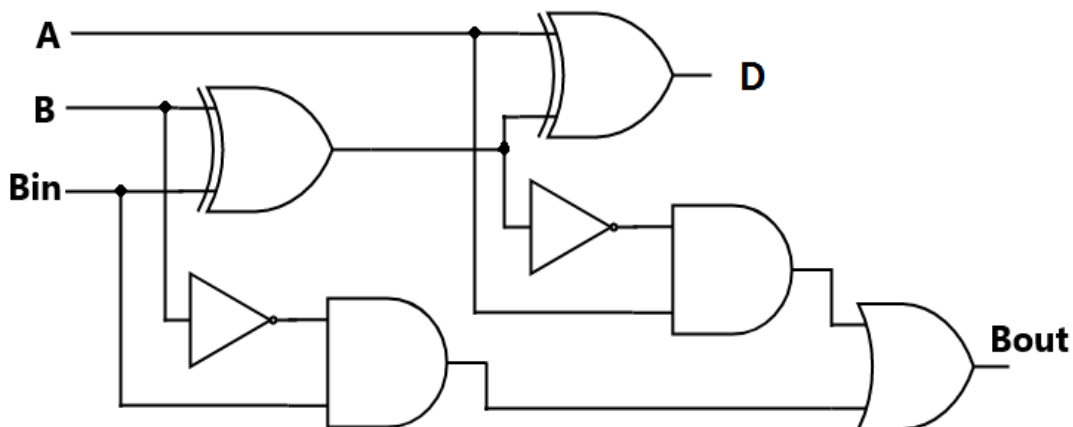
The Boolean expression:

$$D = A \oplus B \oplus B_{in}$$

$$B_{out} = A' B_{in} + A' B + B B_{in}$$

CIRCUIT DAIGRAM:

Truth Table:



Truth Table:

A	B	B_{in}	D	B_{out}
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

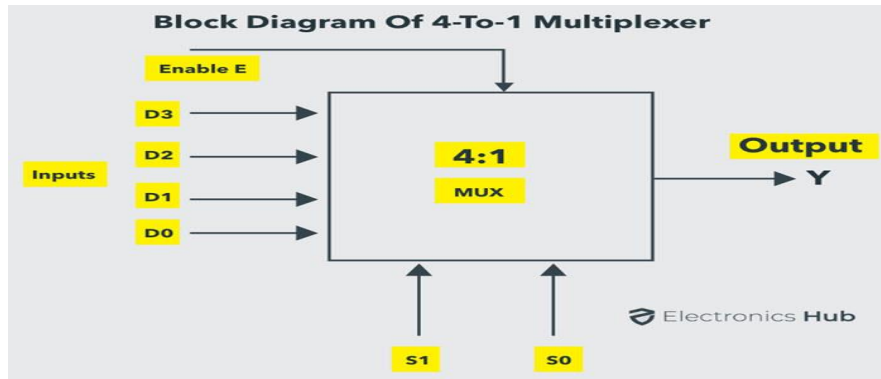
RESULT – Half subtractor and Full subtractor circuit is studied and verified.

EXPERIMENT NO: 6

Aim: - Implementation of 4*1 multiplexer using logic gates.

APPARATUS REQUIRED: Power Supply, Digital Trainer, Connecting Leads, IC's 74153(4x1 multiplexer).

THEORY: A **4-to-1 multiplexer** consists four data input lines as D0 to D3, two select lines as S0 and S1 and a single output line Y. The select lines S0 and S1 select one of the four input lines to connect the output line. The figure below shows the block diagram of a 4-to-1 multiplexer in which, the multiplexer decodes the input through select line.



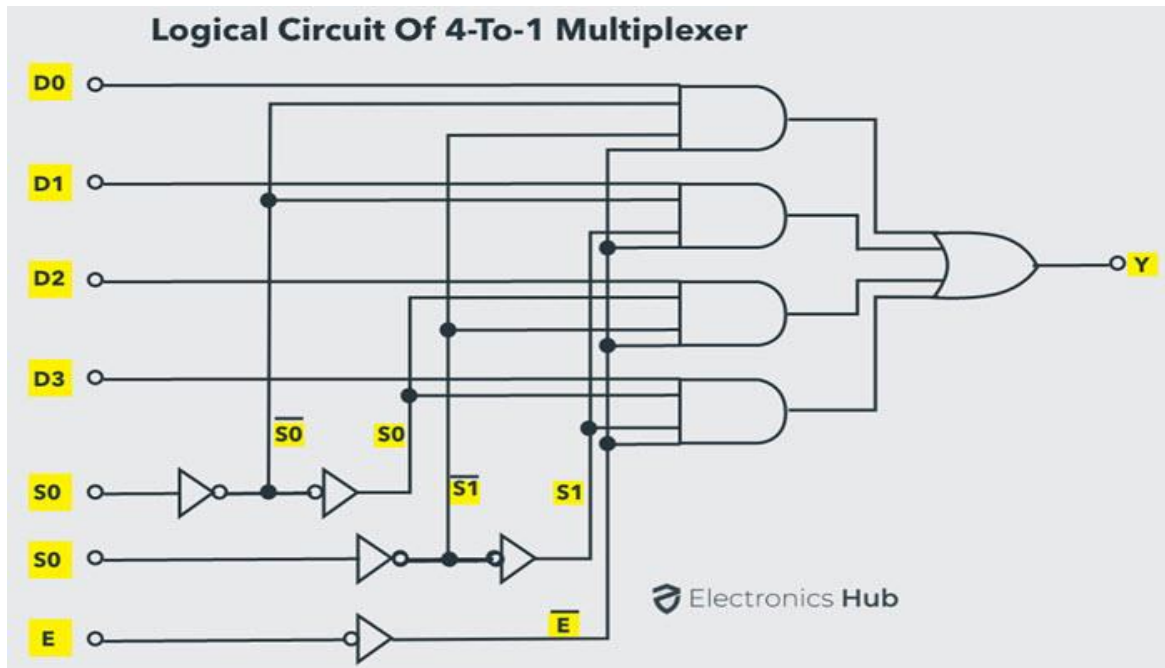
The **truth table** of a 4-to-1 multiplexer is shown below in which four input combinations 00, 10, 01 and 11 on the select lines respectively switches the inputs D0, D2, D1 and D3 to the output. That means when S0=0 and S1 =0, the output at Y is D0, similarly Y is D1 if the select inputs S0=0 and S1= 1 and so on.

S0	S1	D0	D1	D2	D3	Y
0	0	0	X	X	X	0
0	0	1	X	X	X	1
0	1	X	0	X	X	0
0	1	X	1	X	X	1
1	0	X	X	0	X	0
1	0	X	X	1	X	1
1	1	X	X	X	0	0
1	1	X	X	X	1	1

Expressions as follows:

$$Y = \overline{S_0} \overline{S_1} D_0 + \overline{S_0} S_1 D_1 + S_0 \overline{S_1} D_2 + S_0 S_1 D_3$$

A 4-to-1 multiplexer can be implemented by using basic logic gates. The below figure shows the logic circuit of 4:1 MUX which is implemented by four 3-inputs AND gates, two 1-input NOT gates, and one 4-inputs OR gate.



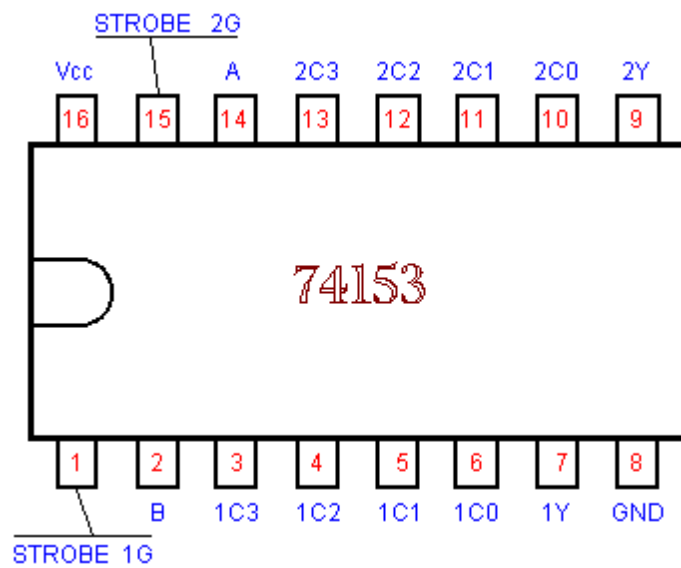
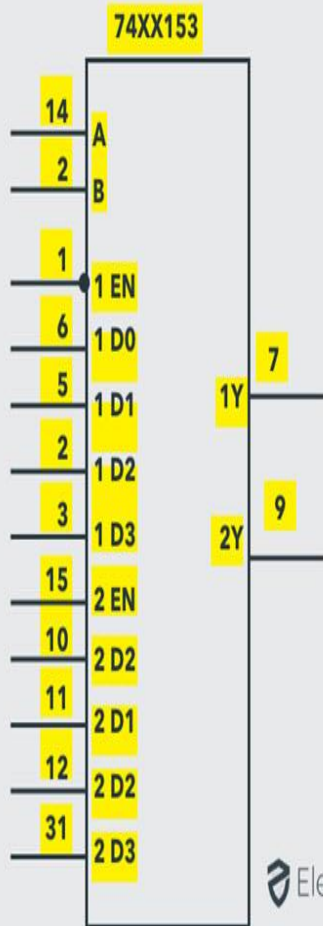
In this circuit, each data input line is connected as input to an AND gate and two select lines are connected as other two inputs to it. Additionally, there is also an Enable Signal. The output of all the AND gates are connected to inputs of OR gate in order to produce the output Y.

CIRCUIT DIAGRAM:

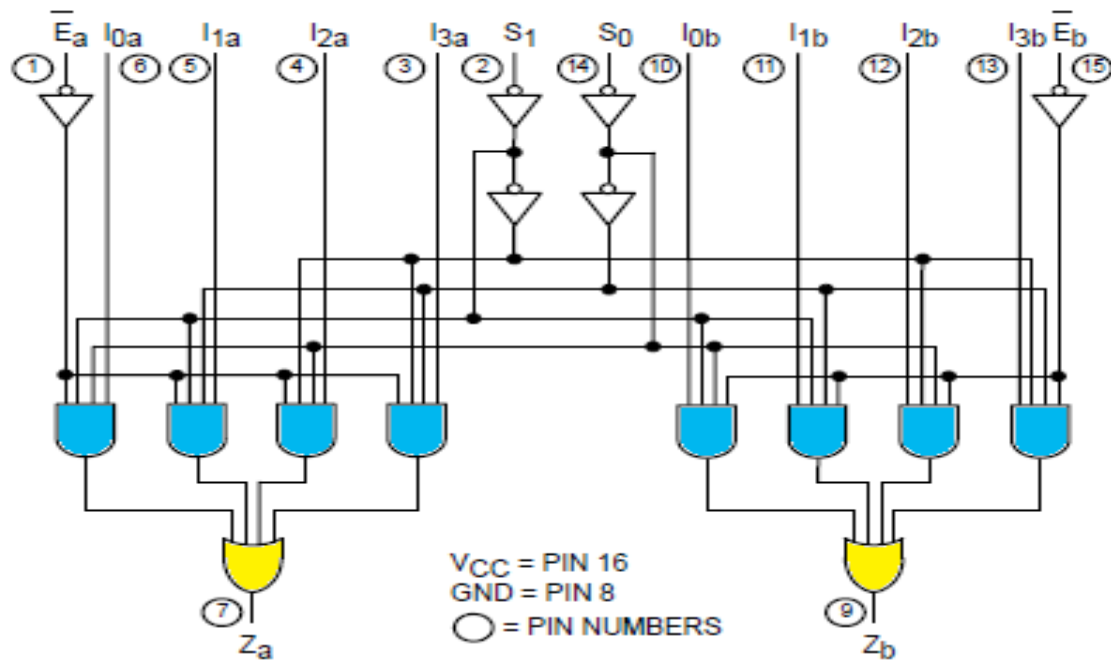
Generally, this type of multiplexers is available in IC with dual mode i.e., there will be two 4-to-1 Multiplexers in a single IC. The most common and popular 4-to-1 line multiplexer is IC 74153 which, is a dual 4-to-1 line multiplexer. It consists of two identical 4-to-1 multiplexers. It has two separate enable or strobe inputs to switch ON or OFF the individual multiplexers. But the Select lines are common to both the Multiplexers.

The figure below shows the pin diagram of IC74153.

Pinout Of Ic 74153



LOGIC DIAGRAM



Detail:

<https://www.electronicshub.org/multiplexerandmultiplexing/#:~:text=The%20most%20common%20and%20popular,%2Dto%2D1%20line%20multiplexer.>

RESULT: Verify the truth table of multiplexer for various inputs.

EXPERIMENT NO: 7

Aim: - Implementation and verification of Decoder/ DE multiplexer and Encoder using logic gates.

EXPERIMENT NO: 8

Aim: - Verification of state table of RS, JK, T and D flip-flops using NAND & NOR gates.

EXPERIMENT NO: 9

Aim: - Design and verify the 4-bit synchronous counter.

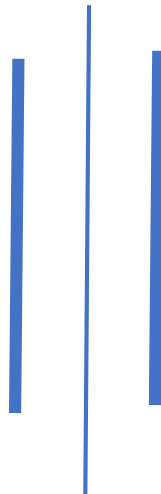
EXPERIMENT NO: 10

Aim: - Design and verify the 4-bit synchronous counter.

Kathmandu Bernhardt College

Tribhuvan University

Bafal, Kathmandu



Digital Logic Lab Report

SUBMITTED BY:

Name:

Roll:

Date:

SUBMITTED TO:

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Department of CSIT**

Internal Teacher's Signature

External's Signature

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Experiment No: 1

AIM: Introduction to digital electronic Lab-nomenclature of digital ICs, specifications, study of the data sheet, concept of VCC and Ground, verification of truth tables of logic gates using TTL ICs in Proteus Simulator.

APPARATUS REQUIRED: Logic State, Logic Probe (Big), Connecting Leads, 74 series IC's.

Gates	IC NO.
AND	7408
OR	7432
NAND	7400
NOR	7402
NOT	7404
XOR	74136

THEORY:

Logic States are the digital components that either give a true or false signal in binary terms that is 1 and 0 respectively.

Logic Probes are the digital components that are used to analyze the logic states of digital signals.

Connecting Leads are the wires used to connect between any digital components with inputs and outputs.

Voltage at the common collector (VCC) is the supply voltage to the circuit.

Ground (GND) is the common return path for electric current.

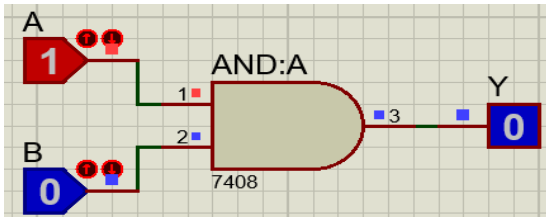
Integrated Circuits are sets of electronic circuits, in this case the logic gates made out of very small transistors and containing usually 4 gates in total except the inverter gates which are 6 in number within the 74 series.

PROCEDURE:

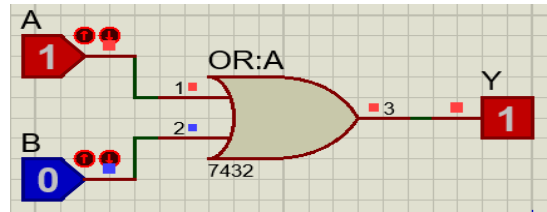
- We fixed the ICs on the board with the help of Device Selector.
- We added Logic States on the board and connected them to the input pins of the gates with connection leads.
- We added Logic Probe on the output pins of the gates and connected them with connecting leads.
- We noted the values of output from the Logic Probe for different combination of inputs from the Logic States and drew the truth table and verified it with the theorized truth tables.

LOGIC DIAGRAMS:

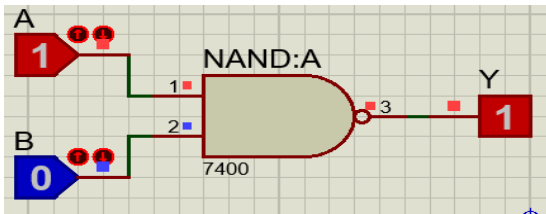
a) AND gate (7408)



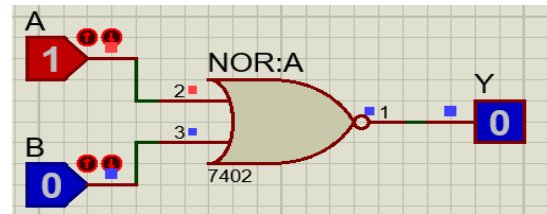
b) OR gate (7432)



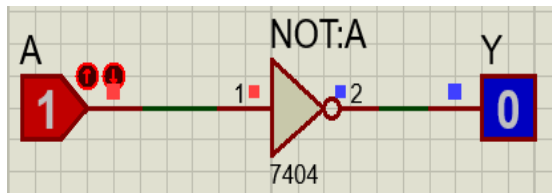
c) NAND gate (7400)



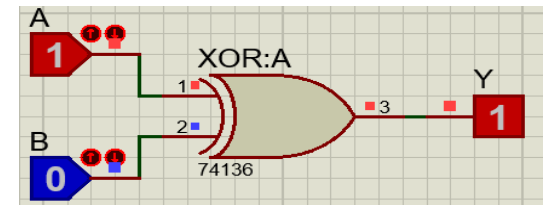
d) NOR gate (7402)



e) NOT gate (7404)



f) XOR gate (74136)



TRUTH TABLES:

a) AND gate (7408)

A(1)	B(2)	Y(3)
0	0	0
0	1	0
1	0	0
1	1	1

b) OR gate (7432)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

c) NAND gate (7400)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

d) NOR gate (7402)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

e) NOT gate (7404)

A(1)	Y(2)
0	1
1	0

f) XOR gate (74136)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

RESULT: All the gates were verified. Observed outputs matched theoretical concepts.

Experiment No: 2

AIM: To study and verify NAND and NOR gates as universal gate in Proteus simulator.

APPARATUS REQUIRED: Logic Probe (Big), Logic State, IC 7400 (NAND Gate), connecting leads.

THEORY: NAND or NOR gate is sufficient for the realization of any logic expression due to this reason, NAND and NOR gates are known as UNIVERSAL gates.

Also proving that the NAND gate is a universal gate proves the NOR as a universal gate upon the realization of duality principle.

LOGIC DIAGRAM:

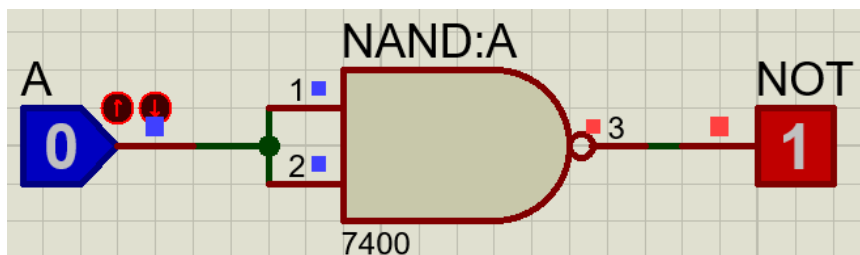


Figure 1: NOT logic operation

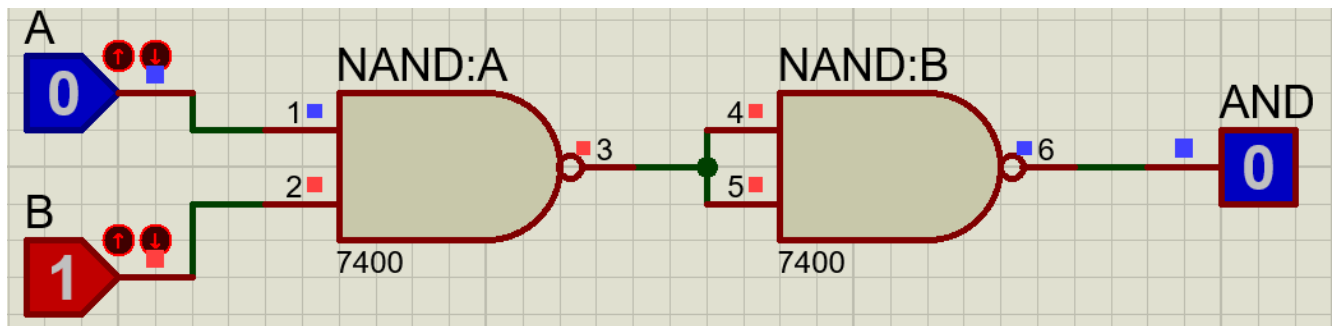


Figure 2: AND logic operation

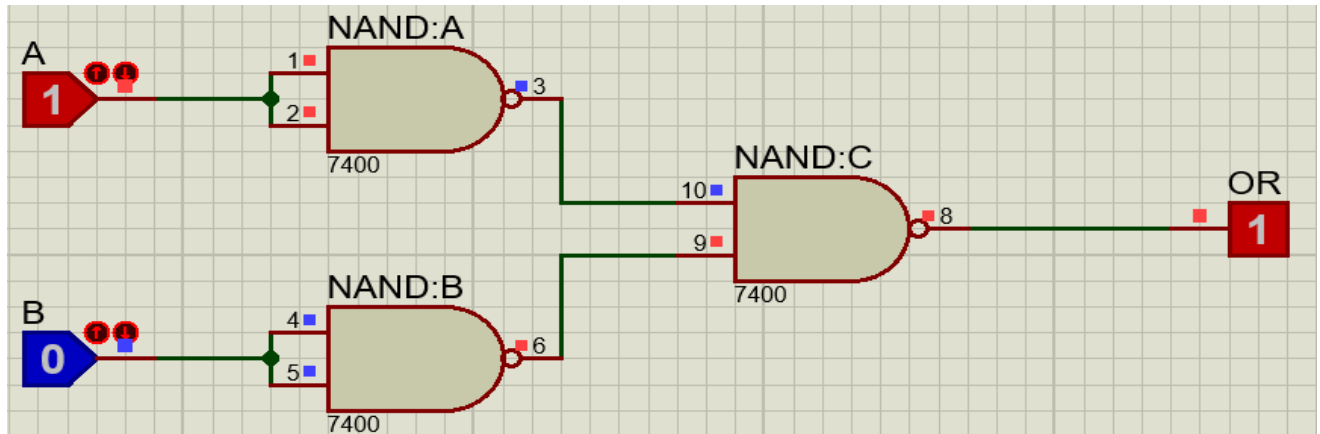


Figure 2: OR logic operation

PROCEDURE:

- We made the connections as per the logic diagram.
- We applied different combination of inputs through the Logic States.
- The Outputs through the Logic Probe were noted.
- We verified the truth table.

TRUTH TABLES:

- a) NOT operation

A(1.2)	Y(3)
0	1
1	0

- b) AND operation

A(1)	B(2)	Y(6)
0	0	0
0	1	0
1	0	0
1	1	1

- c) OR operation

A(1.2)	B(4.5)	Y(8)
0	0	0
0	1	1
1	0	1
1	1	1

RESULT:

All the Universal gates were verified. Observed output matches theoretical concepts.

Experiment No: 3

AIM: Verifying the De Morgan's Theorem.

APPARATUS REQUIRED: Logic Probe (Big), Logic State, Connecting leads, ICs (7404, 7408, 7432, and 7404).

THEORY: According to De Morgan's theorem the following statements must be true

1. $(A + B)' = A' \cdot B'$
2. $(A \cdot B)' = A' + B'$

Proving any one of this will prove the other.

Circuit Diagram:

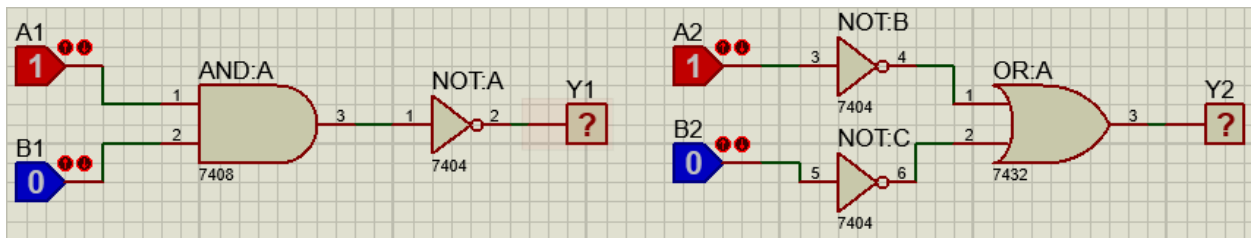


Fig: $(A \cdot B)'$

Fig: $A' + B'$

TRUTH TABLE:

A	B	Y1
0	0	1
0	1	1
1	0	1
1	1	0

A	B	Y2
0	0	1
0	1	1
1	0	1
1	1	0

From the above table $Y1=Y2$, hence they are equivalent to each other

RESULT: The De Morgan's theorem was verified.

Experiment No: 4

AIM: Implementation of the given Boolean function using logic gates in both POS and SOP forms in Proteus simulator.

APPARATUS REQUIRED: Logic Probe (Big), Logic State, Connecting leads, ICs (7404, 7408, 7432, 74HC4075 (triple input OR gate), 7411 (triple input AND gate)).

THEORY: Karnaugh maps are used extensively for the simplification of Boolean functions. It is mostly used for functions having up to 6 variables. These simplified functions yield the same output as the initial functions but with less use of variables and operations.

Example: $f(A,B,C,D) = A'BC + AB'C + ABC' + ABC$ (SOP)'s reduced form is $BC + AC + AB$, and POS form is $f(X,Y,Z) = Y'(X' + Y + Z')(X + Z)$.

LOGIC DIAGRAM:

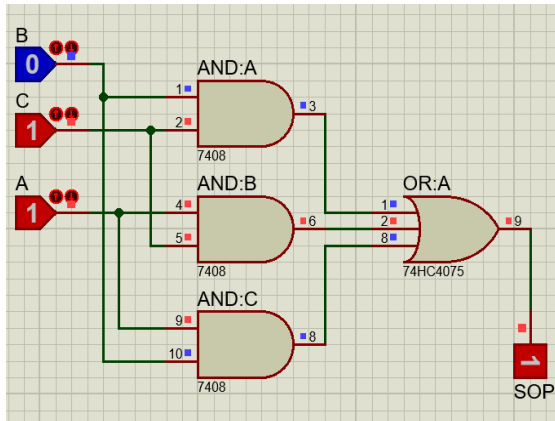


Figure 1: SOP form

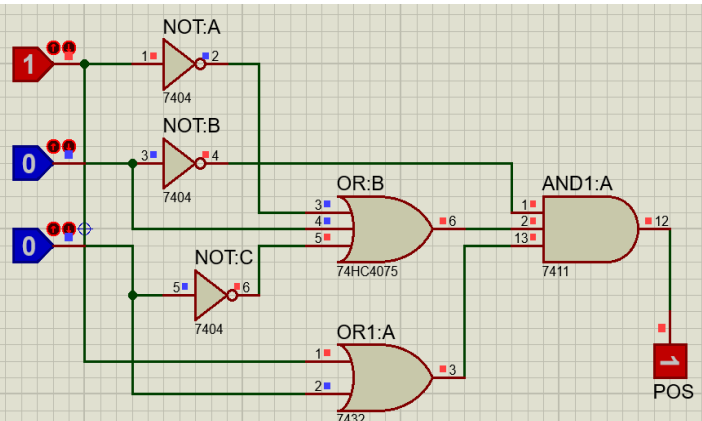


Figure 2: POS form

PROCEDURE:

- With the given equation in SOP/POS form, we first drew the K-map
- We minimized and received a reduced equation from the K-map.
- We made the connections as per the logic diagram.

RESULT:

Implementation of SOP and POS form is obtained with AND and OR gates.

Experiment No: 5

AIM: To design and verify operation of half adder and full adder in Proteus simulator.

APPARATUS REQUIRED: Logic Probe (Big), Logic State, Connecting leads, ICs (74136, 7432 and 7408).

THEORY:

Half adder is a logic circuit that adds two bits. It produces the outputs: sum and carry. The Boolean equation for sum and carry are.

$$\text{SUM} = A \oplus B$$

$$\text{CARRY} = A \cdot B$$

Therefore, sum produces 1 when A & B are different and carry is 1 when A & B are 1.

TRUTH TABLE:

Input		Output	
A	B	SUM	CARRY
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

CIRCUIT DIAGRAM:

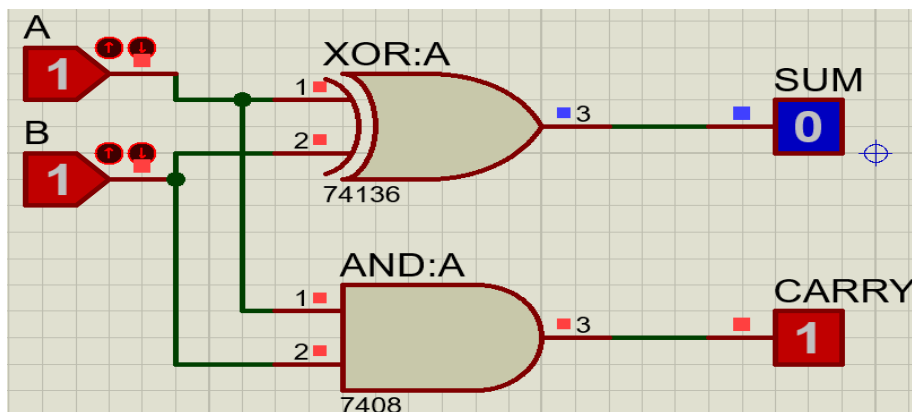


Fig 1: Half Adder

Full Adder is a logic circuit that can add three bits. It produces two outputs: sum and carry. The Boolean equation for sum and Carry are:

$$\text{SUM} = (A \oplus B) \oplus C_{in}$$

$$\text{CARRY} = A \cdot B + C_{in} (A \oplus B)$$

Therefore, sum produces 1 when input is containing odd no's of one and carry is producing 1 when there are two or more 1 in inputs.

TRUTH TABLE:

Input			Output	
A	B	C _{in}	SUM	CARRY
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

CIRCUIT DIAGRAM:

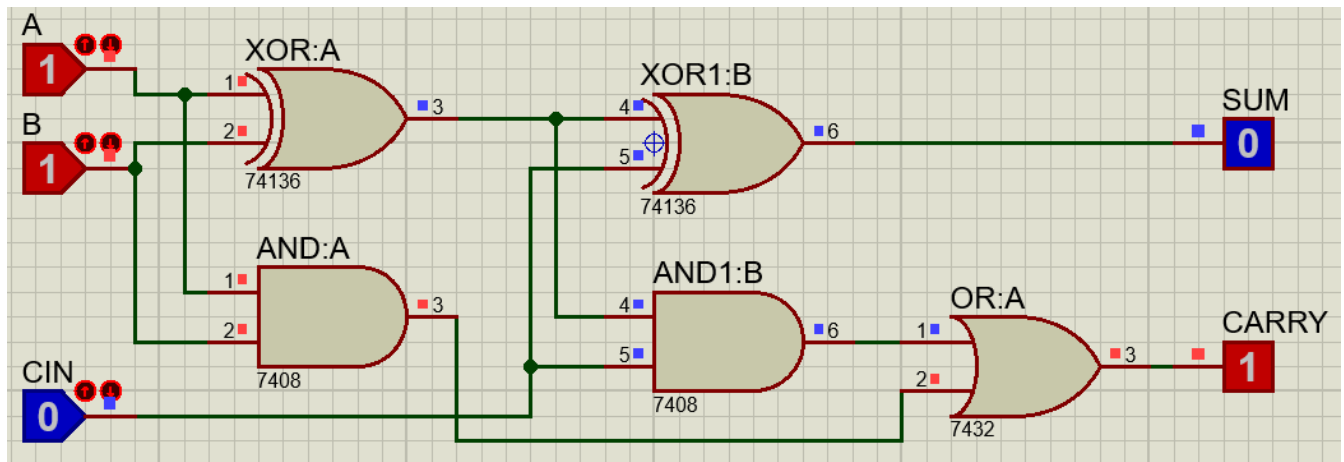


Fig 2: Full adder

RESULT: The half adder and full adder were verified. The outputs coincided the theoretical output.

Experiment No: 6

AIM: To design and verify operation of half subtractor and full subtractor in proteus simulator.

APPARATUS REQUIRED: Logic Probe (Big), Logic State, Connecting leads, ICs (74136, 7408 and 7404).

THEORY:

Half Subtractor is a combinational circuit which is used to perform subtraction of two bits.

It has two inputs, A (Minuend) and B (Subtrahend) and two outputs Difference and borrow:

The Boolean expression for them is:

$$S = A \oplus B$$

$$B_{out} = A' \cdot B$$

TRUTH TABLE:

Input		Output	
A	B	S	B _{out}
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

CIRCUIT DIAGRAM:

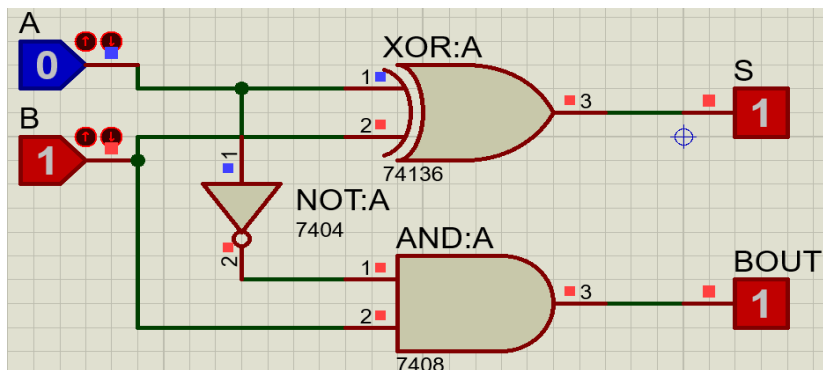


Fig 1: Half subtractor

Full Subtractor is a combinational circuit used to perform subtraction involving 3 bits, namely A (minuend), B (subtrahend), and B_{in} (Borrow bit) and it produces two outputs: Difference (S) and borrow out (B_{out})

$$S = A \oplus B \oplus B_{in}$$

$$B_{out} = A' \cdot B_{in} + A' \cdot B + B \cdot B_{in}$$

TRUTH TABLE:

Input			Output	
A	B	B_{in}	D	B_{out}
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

CIRCUIT DIAGRAM:

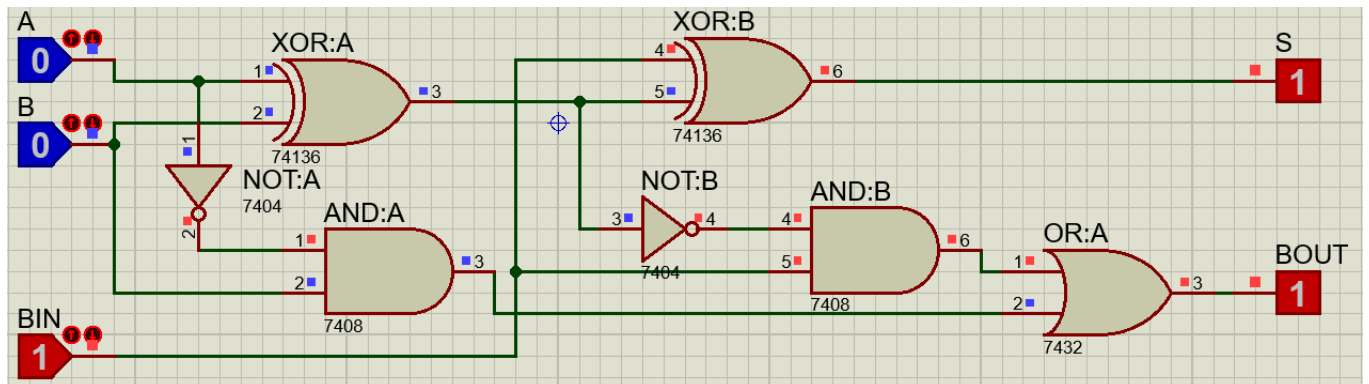


Fig 2: Full Subtractor

RESULT: The half subtractor and full subtractor were verified. The outputs coincided the theoretical output.

Experiment No: 7

AIM: Implementation of 4X1 multiplexer using logic gates.

APPARATUS REQUIRED: Logic Probe (Big), Logic State, Connecting leads, ICs (74153(MUX), 74HC21(quadruple AND gate), 74HC4072(quadruple OR gate) and 7404).

THEORY:

Digital Multiplexer is a combinational circuit that selects binary information from one of many input lines and directs it to a single output line. The selection of a particular line is controlled by a set of selection line. It has 2^n input lines and n selection lines.

A **4-to-1** multiplexer consists of 4 input lines as D₀ to D₃, two selection lines as S₀ and S₁ and a single output line Y.

TRUTH TABLE:

Truth table for the function $F(A,B,C)=\sum(1,3,5,6)$ with the use of 4 to 1 multiplexer is as follows.

E	S ₀	S ₁	D ₀	D ₁	D ₂	D ₃	Y
0	x	x	0	0	0	0	0
1	0	1	0	0	0	0	0
1	0	1	0	1	0	0	1
1	1	0	0	0	A	0	A
1	1	1	0	0	0	A'	A'

Expression as follows:

$$Y=A'.B'.C+A'.B.C+A.B'.C+A.B.C'$$

A 4 to 1 multiplexer can be implemented by using basic logic gates. The figure below shows the logic circuit of 4:1 MUX implemented by four 4-input AND gates, 5 NOT gates, and one 1-input OR gate.

Logical Circuit Diagram of 4 to 1 multiplexer:

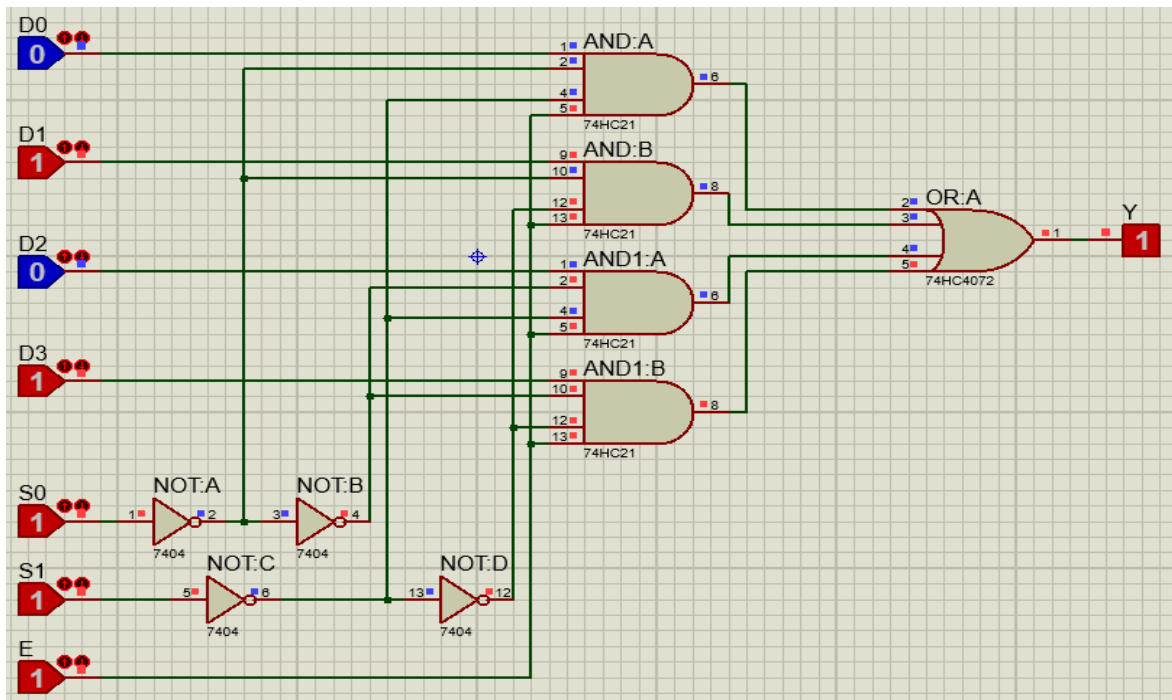


Fig: Logical Circuit Diagram of 4 to 1 multiplexer.

CIRCUIT DIAGRAM:

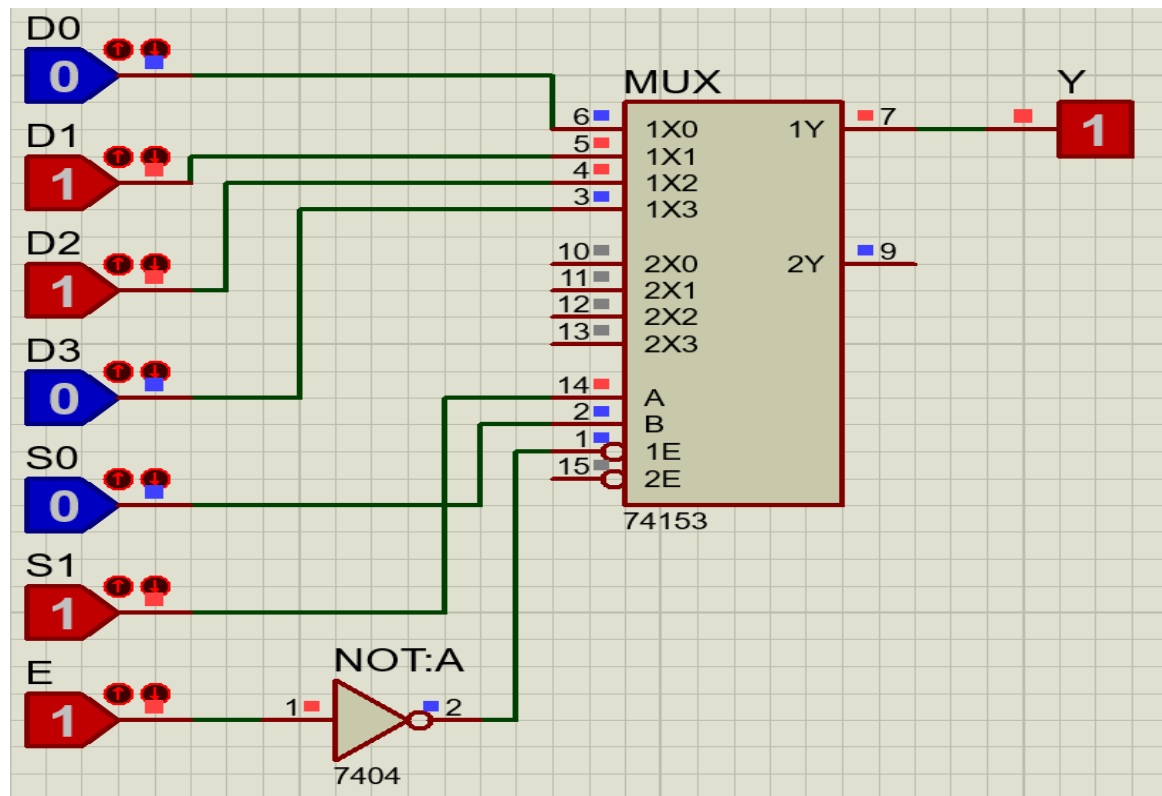


Fig: Circuit Diagram of 4 to 1 multiplexer

RESULT: The output of the multiplexer was verified with the theoretical outputs.

Experiment No: 8

AIM: Implementation of 1X4 demultiplexer using logic gates.

APPARATUS REQUIRED: Logic Probe (Big), Logic State, Connecting leads, ICs (74155(DEMUX), 74HC21(quadruple AND gate), and 7404).

THEORY:

A 1 to 4 demultiplexer is a combinational circuit which receives data from one line (Din) and transmits this data on one of 4 (D0 to D3) output lines.

The selection of output lines is done with a combination of two selection lines A and B.

TRUTH TABLE:

E	S0	S1	Din	D0	D1	D2	D3
0	X	X	1	0	0	0	0
1	0	0	1	1	0	0	0
1	0	1	1	0	1	0	0
1	1	0	1	0	0	1	0
1	1	1	1	0	0	0	1

Here the input line Din is in high state.

Logical Circuit Diagram for 1 to 4 demultiplexer:

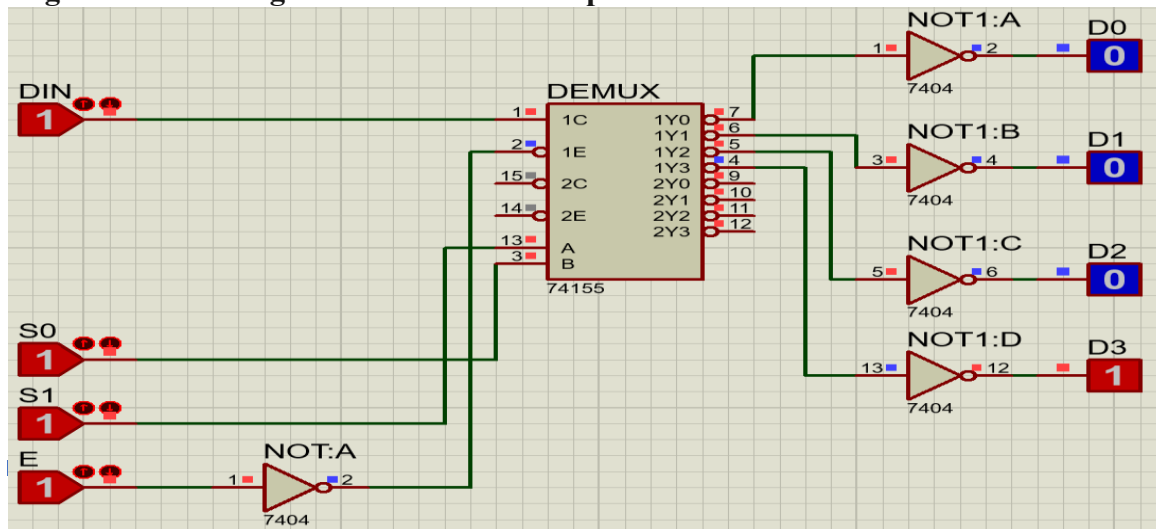


Fig: logical circuit diagram of 1 to 4 demultiplexer

Circuit Diagram:

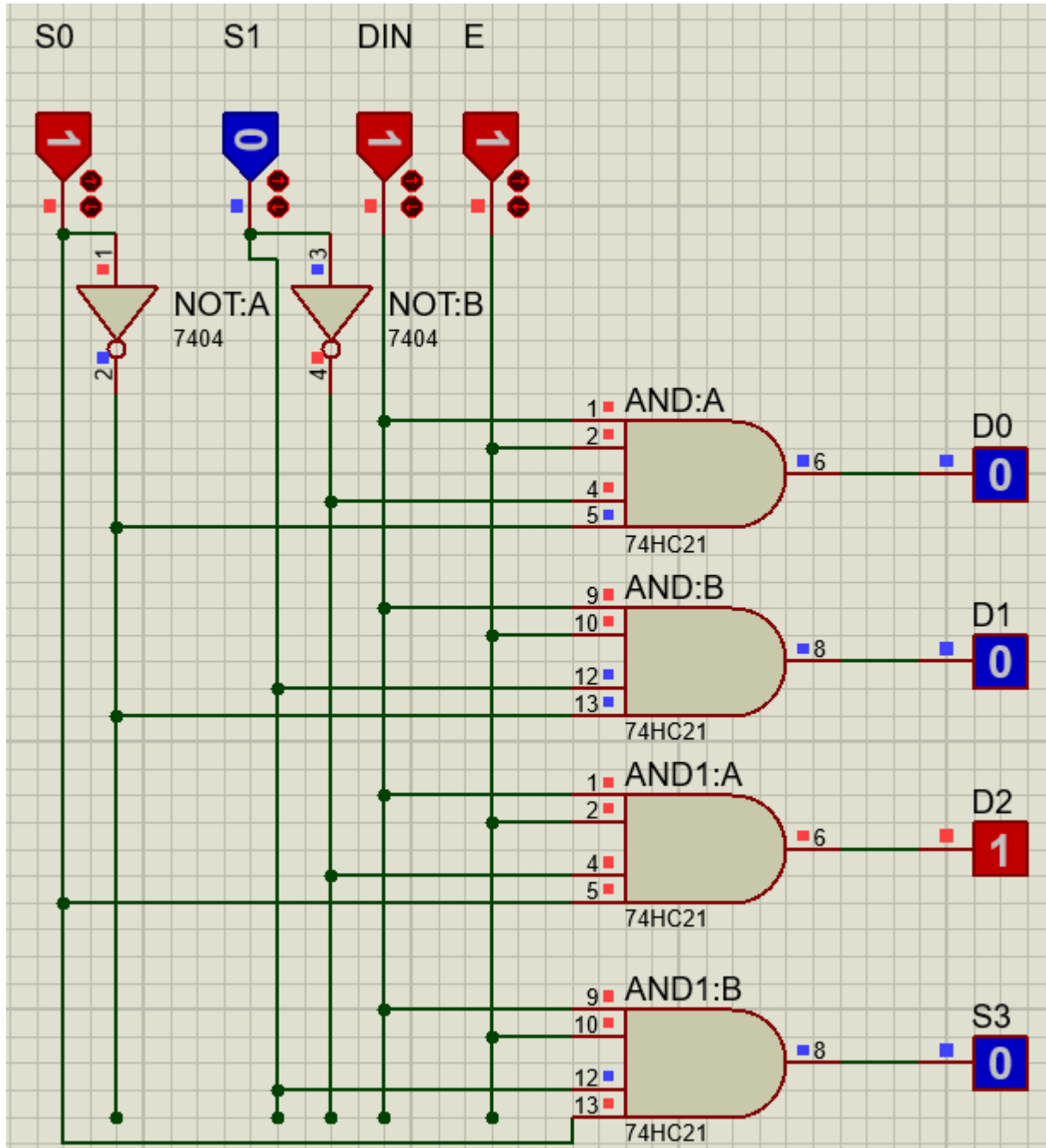


Fig: circuit diagram of 1 to 4 demultiplexer

RESULT: the 1 to 4 demultiplexer was verified.

AIM: Implementation of 3 to 8 decoder using logic gates.

APPARATUS REQUIRED: Logic Probe (Big), Logic State, Connecting leads, ICs (74HC138(DECODER), 74HC4072(quadruple OR gate), and 7404).

THEORY:

A 3 to 8 decoder is a combinational circuit which receives data form n input lines and transmits this data on $m \leq 2^n$ output lines.

The output are in the form of min-terms.

TRUTH TABLE:

The inputs to the decoder are taken through the A, B and C lines and the outputs through 8 lines (D0 to D8). The enable pins are made to enable the decoder and the output is recorded as follows:

E	A	B	C	D0	D1	D2	D3	D4	D5	D6	D7
0	X	X	X	X	X	X	X	X	X	X	X
1	0	0	0	1	0	0	0	0	0	0	0
1	0	0	1	0	1	0	0	0	0	0	0
1	0	1	0	0	0	1	0	0	0	0	0
1	0	1	1	0	0	0	1	0	0	0	0
1	1	0	0	0	0	0	0	1	0	0	0
1	1	0	1	0	0	0	0	0	1	0	0
1	1	1	0	0	0	0	0	0	0	1	0
1	1	1	1	0	0	0	0	0	0	0	1

Logic Circuit Diagram and Circuit Diagram:

Below is the logic circuit diagram of Decoder with the 74HC138 IC. The Enable pins have been set to be enabled when the input through the E state is 1 and the C, B, A correspond to the inputs A, B and C respectively.

The outputs of the decoder are inverted so additional NOT gates are used in conjunction to the Logic probes to get standard output values

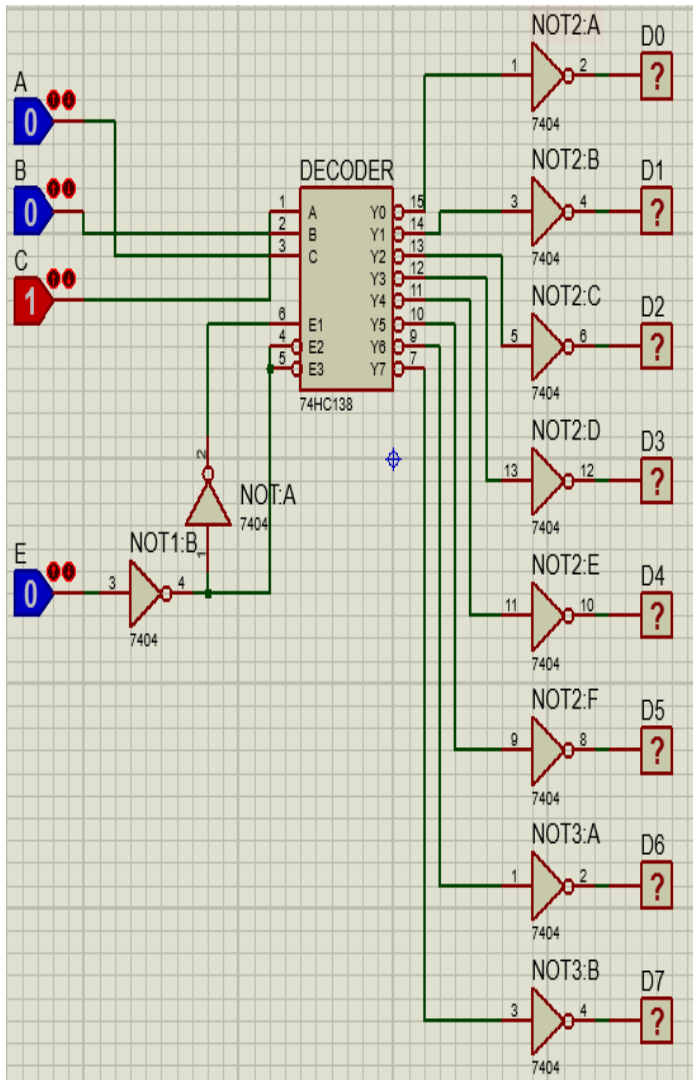


Fig: Logical Circuit Diagram

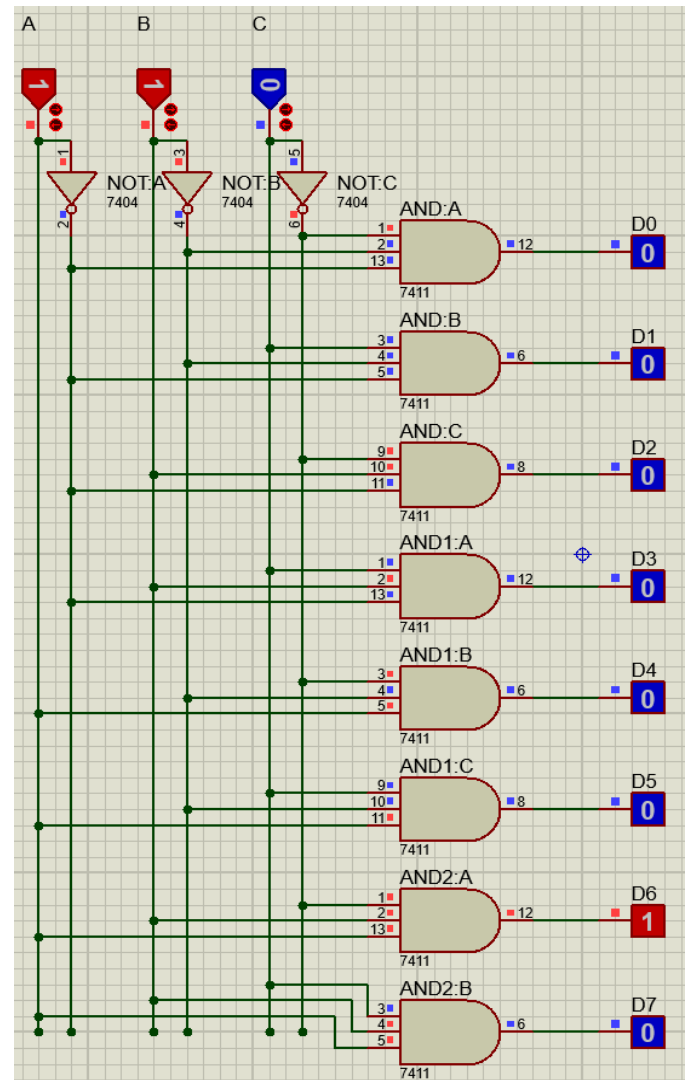


fig: Circuit Diagram

RESULT: The 3 to 8 decoders were verified.

AIM: Implementation of 4 to 2 Encoder using logic gates.

APPARATUS REQUIRED: Logic Probe (Big), Logic State, Connecting leads, ICs (7432).

THEORY:

An **Encoder** is a combination circuit that has a maximum of 2^n input lines and n output lines. Hence it encodes the information from 2^n inputs line on n-bit lines.

A 4 to 2 encoder is an encoder with 4 input lines and 2 output lines.

Circuit Diagram:

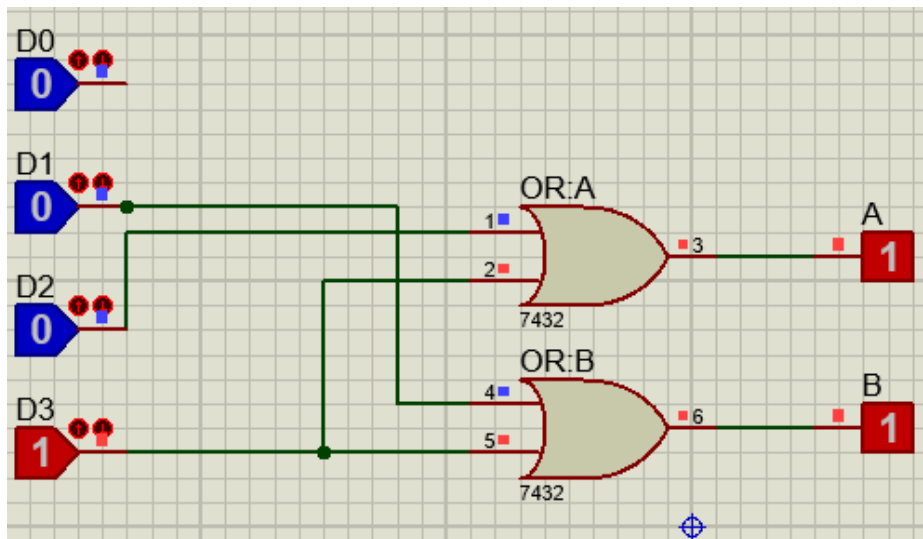


Fig: 4 to 2 Encoder

TRUTH TABLE:

D0	D1	D2	D3	A	B
1	0	0	0	0	0
0	1	0	0	0	1
0	0	1	0	1	0
0	0	0	1	1	1

RESULT: The 4 to 2 encoders were verified.

Experiment No: 9

AIM: Verification of state table of RS flip-flop using NAND and NOR gates.

APPARATUS REQUIRED: Logic Probe (Big), Logic State, Connecting leads, ICs (7400,7402,7408).

THEORY:

RS flip is a flip flop that consists of basic flip flop and two additional NAND gates along with the clock pulse (CP) input. Here CP acts as enable signal and we assume it to be 1 in this case.

STATE TABLE:

Q_n	S	R	Q_{n+1}
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	Indeterminate
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	Indeterminate

CIRCUIT DIAGRAM:

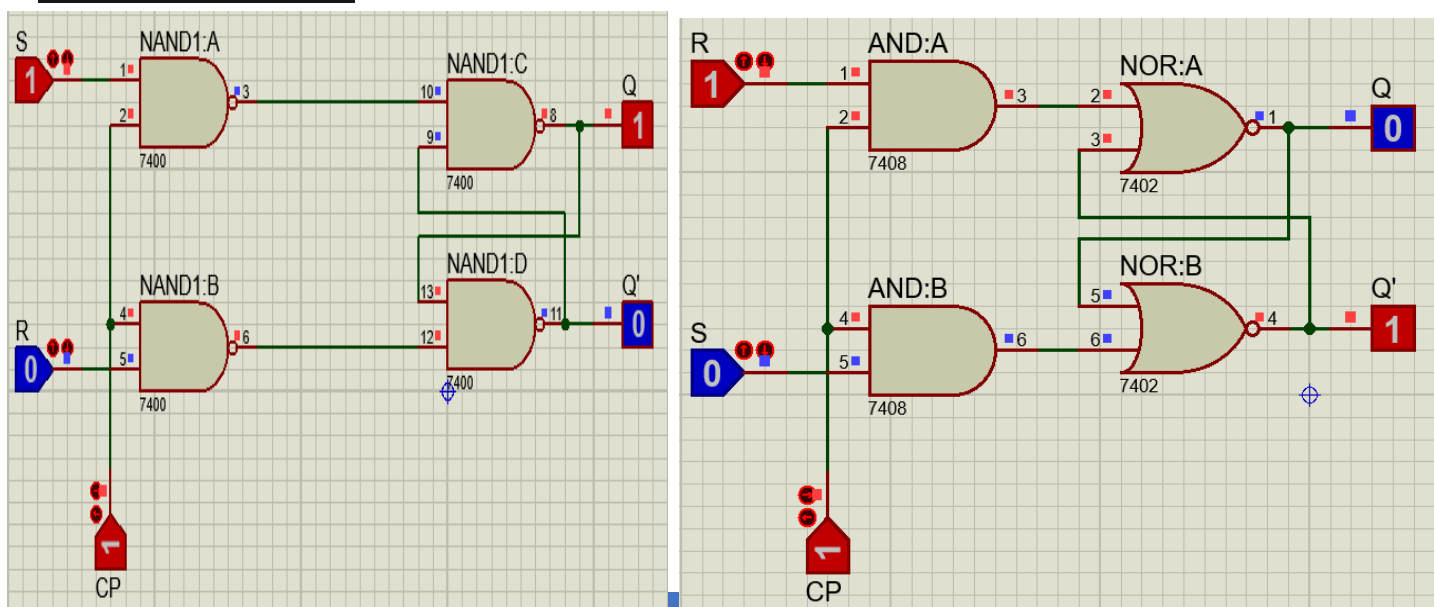


Fig: RS flip flop using NAND gates

Fig: RS flip flop using NOR gates.

RESULT: RS flip-flop was verified.

AIM: Verification of state table of D flip-flop using NAND and NOR gates.

APPARATUS REQUIRED: Logic Probe (Big), Logic State, Connecting leads, ICs (7400,7402,7408,7404).

THEORY:

D flip flop is a flip flop that eliminates the undesirable condition of the indeterminate state in the RS flip-flop by ensuring that the inputs to the basic gates is never equal to '1' or '0' at the same time.

It has two inputs D and CP where CP acts as an enable and is always 1 in this case.

STATE TABLE:

Q_n	S	Q_{n+1}
0	0	0
0	1	1
1	0	0
1	1	1

CIRCUIT DIAGRAM:

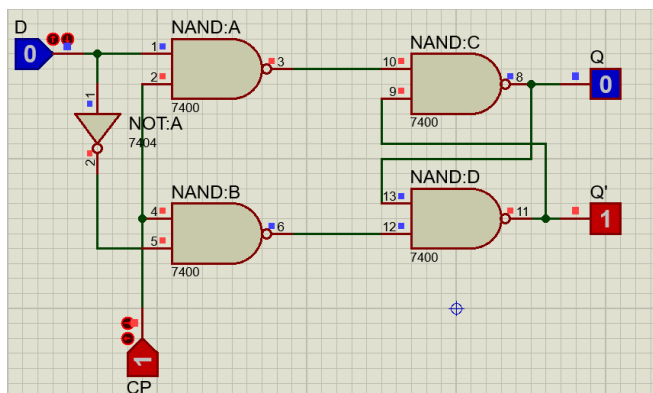


Fig: D flip flop using NAND gates

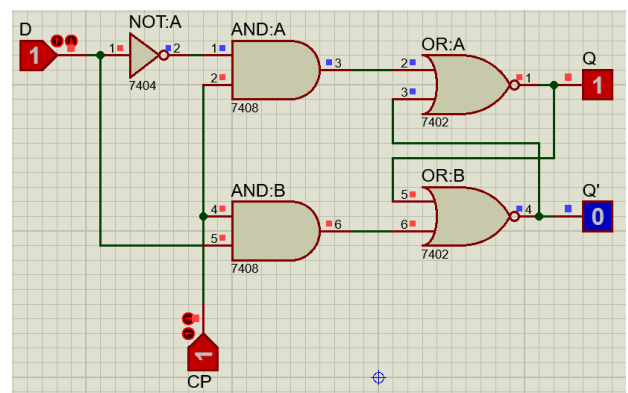


Fig: D flip flop using NOR gates.

RESULT: D flip flops were verified.

AIM: Verification of state table of JK flip-flop using NAND and NOR gates.

APPARATUS REQUIRED: Logic Probe (Big), Logic State, Connecting leads, ICs (7411,7402,7408).

THEORY:

JK flip flop is the refinement of RS flip flop in that the indeterminate state of RS type is defined. Inputs J and K behave like S and R that is Set and Reset respectively and when both J and K are equal, it switches to its complement state.

For JK flip-flop using NAND gate, we make use of extra pins preset and clear because the outputs Q and Q' are initially always simulated in the forbidden state 1. To circumvent this, we run the experiment initially with all the signals 0, then we press Preset and following that we press Clear. This will put the outputs in Q=0 and Q'=1 state and we can then run the following experiments with alternating clock pulses

STATE TABLE:

Q_n	S	R	Q_{n+1}
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

CIRCUIT DIAGRAM:

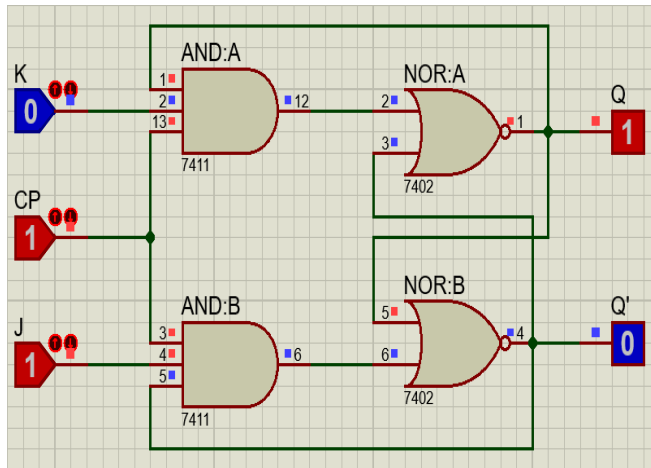


Fig: JK flip flop

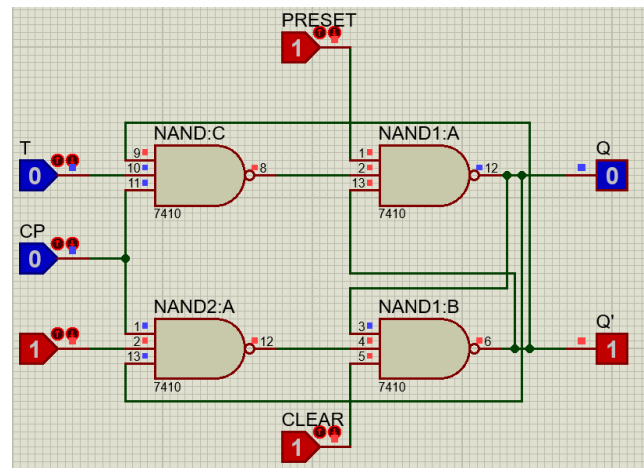


fig: JK flip-flop using NAND gates

RESULT: The JK flip flop was verified.

AIM: Verification of state table of T flip-flop.

APPARATUS REQUIRED: Logic Probe (Big), Logic State, Connecting leads, ICs (7411,7402,7408).

THEORY:

T flip flop is the single input version of JK flip flop and is obtained from the JK flip-flop when the inputs are tied together. The T refers to toggle because of its ability to complement the state.

In this lab we make use of extra pins preset and clear because the outputs Q and Q' are initially always simulated in the forbidden state 1. To circumvent this, we run the experiment initially with all the signals 0, then we press Preset and following that we press Clear. This will put the outputs in Q=0 and Q'=1 state and we can then run the following experiments with alternating clock pulses.

STATE TABLE:

Q_n	S	Q_{n+1}
0	0	0
0	1	1
1	0	1
1	1	0

CIRCUIT DIAGRAM:

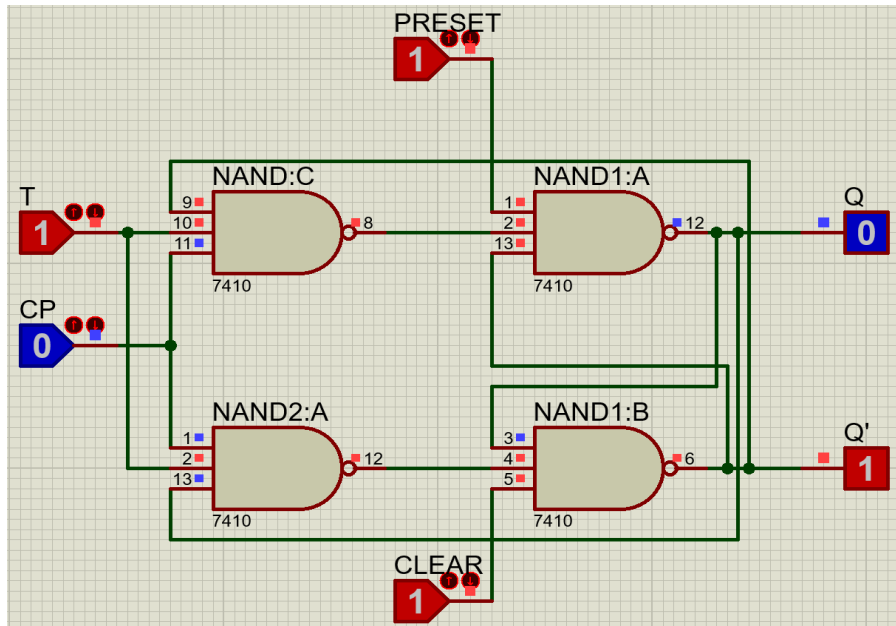


Fig: T flip flop

RESULT: T flip flop was verified.

Experiment No: 10

AIM: Design and verify the 4-bit synchronous counter.

APPARATUS REQUIRED: Logic Probe (Big), Logic State, Connecting leads, ICs (7476 (JK flip flop), 7408).

THEORY:

Counter is a sequential circuit that goes through a predetermined sequence of states upon the application of clock pulses.

Synchronous counter is a counter where clock pulses are applied to all flip-flops, which means that all flip-flops share the same clock signal and output will change at the same time.

DESIGNING 4-BIT SYNCHRONOUS COUNTER:

State Diagram:

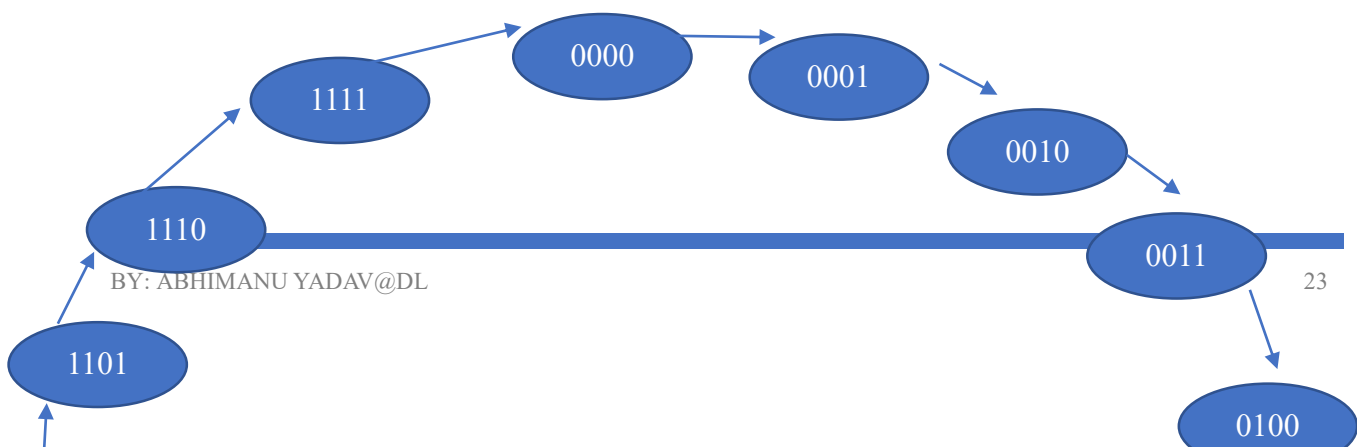


Fig: State Diagram for 4-bit synchronous counter

Excitation Table for JK Flip Flop:

Q_t	Q_{t+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Here, 4 flip flop are required of JK type.

Transition Table:

Present state				Next State				Input JK							
A	B	C	D	A	B	C	D	J_A	K_A	J_B	K_B	J_C	K_C	J_D	K_D
0	0	0	0	0	0	0	1	0	X	0	X	0	X	1	X
0	0	0	1	0	0	1	0	0	X	0	X	1	X	X	1
0	0	1	0	0	0	1	1	0	X	0	X	X	0	1	X
0	0	1	1	0	1	0	0	0	X	1	X	X	1	X	1
0	1	0	0	0	1	0	1	0	X	X	0	0	X	1	X
0	1	0	1	0	1	1	0	0	X	X	0	1	X	X	1
0	1	1	0	0	1	1	1	0	X	X	0	X	0	1	X
0	1	1	1	1	0	0	0	1	X	X	1	X	1	X	1
1	0	0	0	1	0	0	1	X	0	0	X	0	X	1	X

1	0	0	1	1	0	1	0	X	0	0	X	1	X	X	1
1	0	1	0	1	0	1	1	X	0	0	X	X	0	1	X
1	0	1	1	1	1	0	0	X	0	1	X	X	1	X	1
1	1	0	0	1	1	0	1	X	0	X	0	0	X	1	X
1	1	0	1	1	1	1	0	X	0	X	0	1	X	X	1
1	1	1	0	1	1	1	1	X	0	X	0	X	0	1	X
1	1	1	1	0	0	0	0	X	1	X	1	X	1	X	1

K-Mapping:

AB\CD	00	01	11	10
00				
01			1	
11	X	X	X	X
10	X	X	X	X

$$J_A = BCD$$

AB\CD	00	01	11	10
00	X	X	X	X
01	X	X	X	X
11			1	
10				

$$K_A = BCD$$

AB\CD	00	01	11	10
00			1	
01	X	X	X	X
11	X	X	X	X
10			1	

$$J_B = CD$$

AB\CD	00	01	11	10
00	X	X	X	X
01			1	
11			1	
10	X	X	X	X

$$K_B = CD$$

AB\CD	00	01	11	10
00		1	X	X
01		1	X	X
11		1	X	X
10		1	X	X

$$J_C = D$$

AB\CD	00	01	11	10
00	X	X	1	
01	X	X	1	
11	X	X	1	
10	X	X	1	

$$K_C = D$$

AB\CD	00	01	11	10
00	1	X	X	1
01	1	X	X	1
11	1	X	X	1
10	1	X	X	1

$J_D=1$

AB\CD	00	01	11	10
00	X	1	1	X
01	X	1	1	X
11	X	1	1	X
10	X	1	1	X

$K_D=1$

CIRCUIT DIAGRAM:

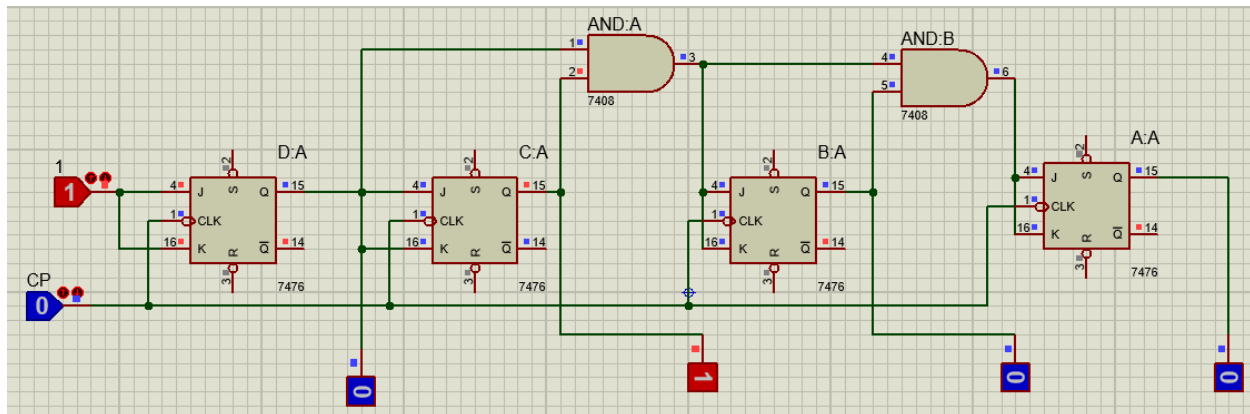


Fig: 4-bit synchronous counter

RESULT: 4-bit synchronous counter was verified.

Experiment No: 11

AIM: Design and verify the 4-bit asynchronous counter.

APPARATUS REQUIRED: Logic Probe (Big), Logic State, Connecting leads, ICs (7476 (JK flip flop), 7408).

THEORY:

Counter is a sequential circuit that goes through a predetermined sequence of states upon the application of clock pulses.

Asynchronous counter is a counter where clock pulses are only applied to the first flip-flop and each subsequent flip-flop is clocked by the preceding flip-flop.

STATE DIAGRAM:

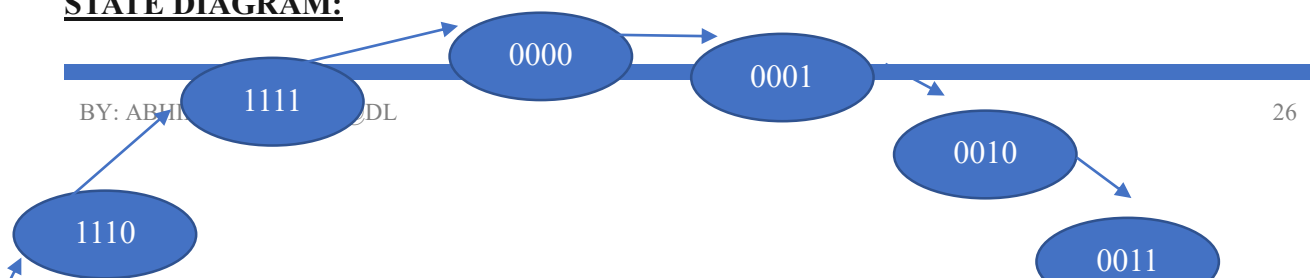


Fig: 4-bit asynchronous counter state diagram

STATE TABLE:

Present state				Next State			
A	B	C	D	A	B	C	D
0	0	0	0	0	0	0	1
0	0	0	1	0	0	1	0
0	0	1	0	0	0	1	1
0	0	1	1	0	1	0	0
0	1	0	0	0	1	0	1
0	1	0	1	0	1	1	0
0	1	1	0	0	1	1	1
0	1	1	1	1	0	0	0
1	0	0	0	1	0	0	1
1	0	0	1	1	0	1	0

1	0	1	0	1	0	1	1
1	0	1	1	1	1	0	0
1	1	0	0	1	1	0	1
1	1	0	1	1	1	1	0
1	1	1	0	1	1	1	1
1	1	1	1	0	0	0	0

CIRCUIT DIAGRAM:

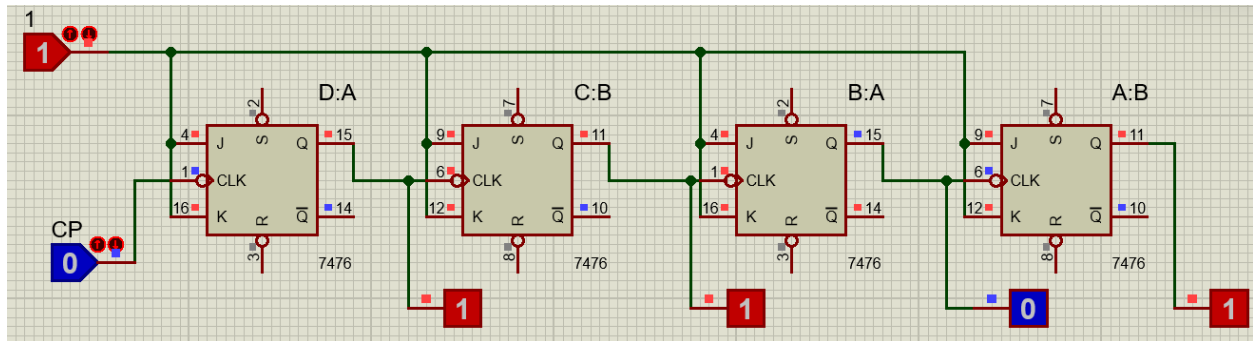


Fig: 4-bit asynchronous counter

RESULT: 4-bit asynchronous counter was verified.